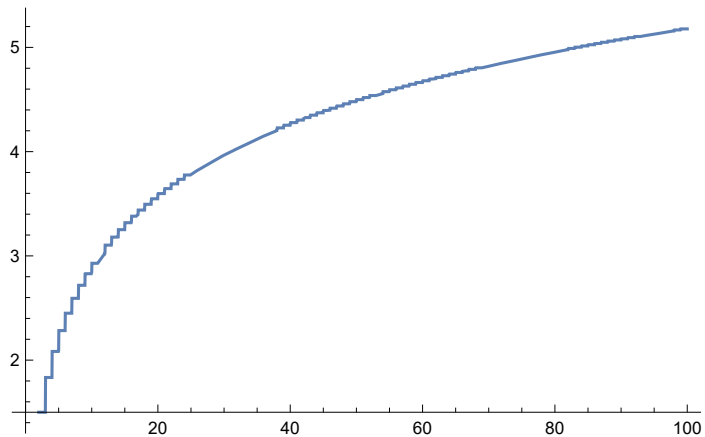


In[]:= (*1*)

Plot[Sum[$\frac{1}{n}$, {n, m}], {m, 2, 100},
[绘图](#) [求和](#)

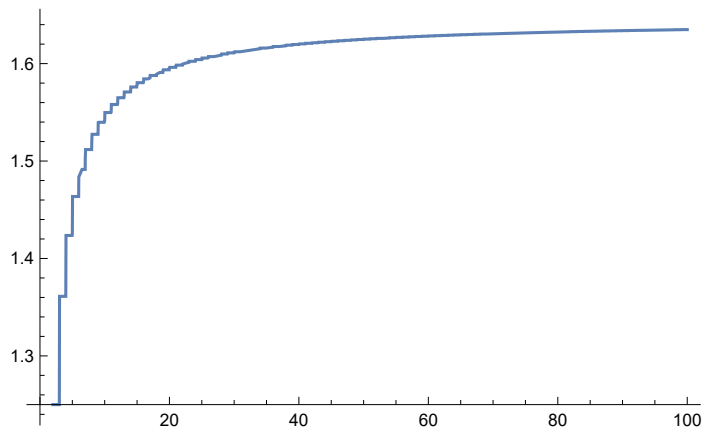
PlotRange → All]
[绘制范围](#) [全部](#)

Out[]:=



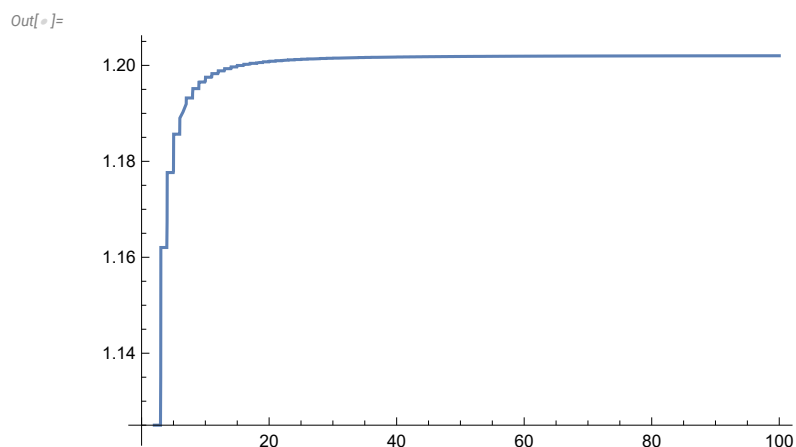
In[]:= Plot[Sum[$\frac{1}{n^2}$, {n, m}], {m, 2, 100}, PlotRange → All]
[绘图](#) [求和](#) [绘制范围](#) [全部](#)

Out[]:=



```
In[ ]:= Plot[Sum[ $\frac{1}{n^3}$ , {n, m}], {m, 2, 100}, PlotRange -> All]
```

绘图 求和 绘制范围 全部



```
In[ ]:= (*2*)
Clear["`*"]
清除
f[x_] := ArcTan[x];
反正切
(*在x=1处展开*)
g[n_] := Normal[Series[f[x], {x, 1, n}]] /. x -> 2
转换... 级数
data = Table[{n, N[g[n]], N[f[2]], N[Abs[f[2] - g[n]] / g[n]] * 100}, {n, 1, 9, 1}];
表格 数值运算 数值运算 绝对值
TableForm[data, TableSpacing -> {1, 5},
表格形式 表格空格
TableHeadings -> {None, {"展开级数n", "在x=1处展开的幂级数的值", "f(1)", "Error(%)"}}]
表格标头 无
```

Out[]//TableForm=

展开级数n	在x=1处展开的幂级数的值	f(1)	Error(%)
1	1.2854	1.10715	13.8673
2	1.0354	1.10715	6.92975
3	1.11873	1.10715	1.03535
4	1.11873	1.10715	1.03535
5	1.09373	1.10715	1.22674
6	1.11456	1.10715	0.665382
7	1.10564	1.10715	0.136795
8	1.10564	1.10715	0.136795
9	1.10911	1.10715	0.176697

```

In[ ]:= (*在x=2处展开*)
g[n_] := Normal[Series[f[x], {x, 2, n}]] /. x -> 3
      [转换... 级数]
data = Table[{n, N[g[n]], N[f[3]], N[Abs[f[3] - g[n]] / g[n]] * 100}, {n, 1, 9, 1}];
      [表格 数值运算 数值运算 ... 绝对值]
TableForm[data, TableSpacing -> {1, 5},
      [表格形式 表格空格]
      TableHeadings -> {None, {"展开级数n", "在x=2处展开的幂级数的值", "f(3)", "Error(%)"}},
      [表格标头 无]

```

Out[]//TableForm=

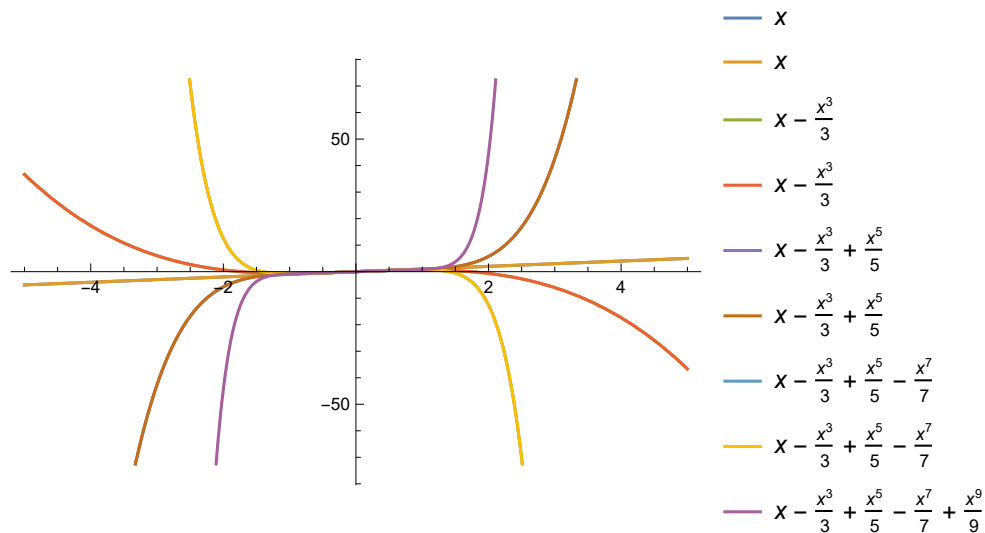
展开级数n	在x=2处展开的幂级数的值	f(3)	Error(%)
1	1.30715	1.24905	4.44501
2	1.22715	1.24905	1.78438
3	1.25648	1.24905	0.591833
4	1.24688	1.24905	0.173531
5	1.24951	1.24905	0.0368369
6	1.24904	1.24905	0.000724927
7	1.24898	1.24905	0.0049707
8	1.24909	1.24905	0.00363759
9	1.24902	1.24905	0.00182326

```

In[ ]:= Plot[Evaluate[Table[Normal[Series[f[x], {x, 0, n}]], {n, 9}]],
      [绘图 计算 表格 转换... 级数]
      {x, -5, 5}, PlotLegends -> "Expressions"]
      [绘图的图例]

```

Out[]:=



```

Clear["`*"]
清除
f[x_] := ArcCos[x];
反余弦
(*在x=0处展开*)
g[n_] := Normal[Series[f[x], {x, 0, n}]] /. x -> 0.01
转换... 级数
data = Table[{n, N[g[n]], N[f[0.01]], N[Abs[f[0.01] - g[n]] / g[n] * 100]}, {n, 1, 9, 1}];
表格 数值运算 数值运算 ... 绝对值
TableForm[data, TableSpacing -> {1, 5},
表格形式 表格空格
TableHeadings -> {None, {"展开级数n", "在x=0处展开的幂级数的值", "f(0.01)", "Error(%)"}}]
表格标头 无

```

Out[]//TableForm=

展开级数n	在x=0处展开的幂级数的值	f(0.01)	Error
1	1.5608	1.5608	0.0000106788
2	1.5608	1.5608	0.0000106788
3	1.5608	1.5608	4.80552×10^{-10}
4	1.5608	1.5608	4.80552×10^{-10}
5	1.5608	1.5608	2.84527×10^{-14}
6	1.5608	1.5608	2.84527×10^{-14}
7	1.5608	1.5608	0.
8	1.5608	1.5608	0.
9	1.5608	1.5608	0.

```

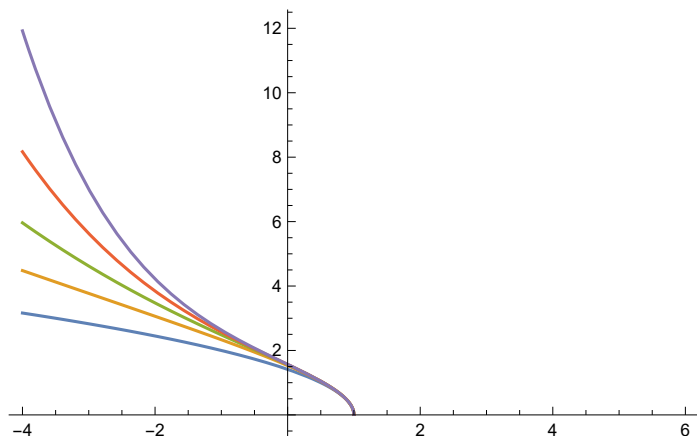
In[ ]:= (*在x=1处展开*)
f[x_] := ArcCos[x];
      |反余弦
g[n_] := Normal[Series[f[x], {x, 1, n}]] /. x -> 1.01
      |转换... |级数
data = Table[{n, N[g[n]], N[f[1.01]], N[Abs[f[1.01] - g[n]] / g[n]] * 100}, {n, 1, 9, 1}];
      |表格 |数值运算 |数值运算 |... |绝对值
TableForm[data, TableSpacing -> {1, 5},
      |表格形式 |表格空格
      TableHeadings -> {None, {"展开级数n", "在x=1处展开的幂级数的值", "f(1.01)", "Error(%)"}},
      |表格标头 |无
Plot[Evaluate[Table[Normal[Series[f[x], {x, 1, n}]], {n, 5}]],
      |绘图 |计算 |表格 |转换... |级数
      {x, -4, 6}, PlotLegends -> "Expressions"]
      |绘图的图例

```

Out[]:=TableForm=

展开级数n	在x=1处展开的幂级数的值	f(1.01)	Error(%)
1	0. + 0.141421 i	0. + 0.141304 i	0. - 0.0831464 i
2	0. + 0.141304 i	0. + 0.141304 i	0. - 0.0001871 i
3	0. + 0.141304 i	0. + 0.141304 i	0. - 5.56607 × 10 ⁻⁷ i
4	0. + 0.141304 i	0. + 0.141304 i	0. - 1.89348 × 10 ⁻⁹ i
5	0. + 0.141304 i	0. + 0.141304 i	0. - 6.99273 × 10 ⁻¹² i
6	0. + 0.141304 i	0. + 0.141304 i	0. - 1.96425 × 10 ⁻¹⁴ i
7	0. + 0.141304 i	0. + 0.141304 i	0. + 0. i
8	0. + 0.141304 i	0. + 0.141304 i	0. + 0. i
9	0. + 0.141304 i	0. + 0.141304 i	0. - 1.96425 × 10 ⁻¹⁴ i

Out[]:=



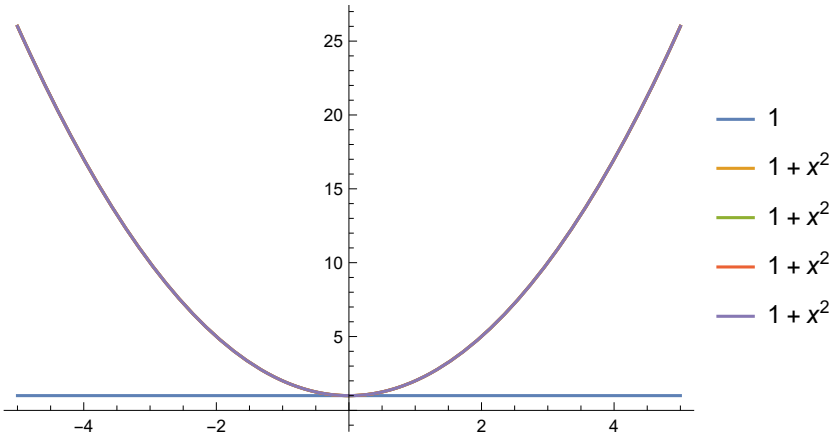
$$\begin{aligned}
 & \text{--- } \frac{\pi}{2} + \frac{1}{2} \left(-\pi + 2i(-1)^{\left\lfloor \frac{-\arg(-1+x)}{2\pi} \right\rfloor} \right) \sqrt{2} \sqrt{-1+x} \\
 & \text{--- } \frac{\pi}{2} + \frac{1}{2} \left(-\pi + (-1)^{\left\lfloor \frac{-\arg(-1+x)}{2\pi} \right\rfloor} \right) \left(2i \sqrt{2} \sqrt{-1+x} \right. \\
 & \text{--- } \frac{\pi}{2} + \frac{1}{2} \left(-\pi + (-1)^{\left\lfloor \frac{-\arg(-1+x)}{2\pi} \right\rfloor} \right) \left(2i \sqrt{2} \sqrt{-1+x} \right. \\
 & \text{--- } \frac{\pi}{2} + \frac{1}{2} \left(-\pi + (-1)^{\left\lfloor \frac{-\arg(-1+x)}{2\pi} \right\rfloor} \right) \left(2i \sqrt{2} \sqrt{-1+x} \right. \\
 & \text{--- } \frac{\pi}{2} + \frac{1}{2} \left(-\pi + (-1)^{\left\lfloor \frac{-\arg(-1+x)}{2\pi} \right\rfloor} \right) \left(2i \sqrt{2} \sqrt{-1+x} \right.
 \end{aligned}$$

```
In[ ]:= Clear["`*"]
清除
(*在x=0处展开*)
f[x_] := Exp[x^2] Cos[x^2];
指数形式 余弦
g[n_] := Normal[Series[f[x], {x, 0, n}]] /. x -> 0.01
转换... 级数
data = Table[{n, N[g[n]], N[f[0.01]], N[Abs[f[0.01] - g[n]] / g[n] * 100]}, {n, 1, 9, 1}];
表格 数值运算 数值运算 ... 绝对值
TableForm[data, TableSpacing -> {1, 5}, TableHeadings ->
表格形式 表格空格 表格标头
{None, {"展开级数n", "在x=0处展开的幂级数的值", "f(0.01)", "Error(%)", }}]
无
Plot[Evaluate[Table[Normal[Series[f[x], {x, 0, n}]], {n, 5}]],
绘图 计算 表格 转换... 级数
{x, -5, 5}, PlotLegends -> "Expressions"]
绘图的图例
```

Out[]//TableForm=

展开级数n	在x=0处展开的幂级数的值	f(0.01)	Error(%)
1	1.	1.0001	0.01
2	1.0001	1.0001	3.33256×10^{-11}
3	1.0001	1.0001	3.33256×10^{-11}
4	1.0001	1.0001	3.33256×10^{-11}
5	1.0001	1.0001	3.33256×10^{-11}
6	1.0001	1.0001	0.
7	1.0001	1.0001	0.
8	1.0001	1.0001	0.
9	1.0001	1.0001	0.

Out[]:=



```

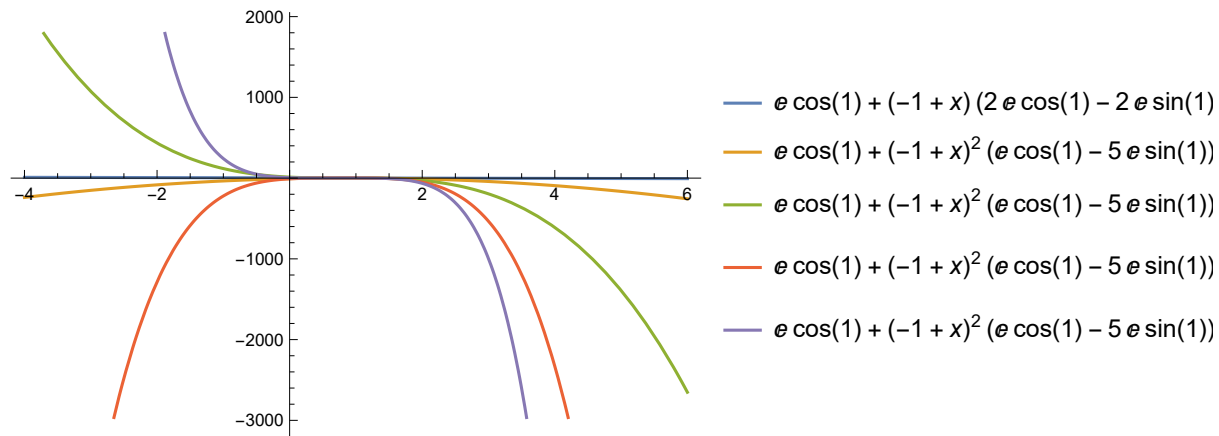
In[ ]:= (*在x=1处展开*)
f[x_] := Exp[x^2] Cos[x^2];
      |指数形式 |余弦
g[n_] := Normal[Series[f[x], {x, 1, n}]] /. x -> 1.01
      |转换... |级数
data = Table[{n, N[g[n]], N[f[1.01]], N[Abs[f[1.01] - g[n]] / g[n] * 100]}, {n, 1, 9, 1}];
      |表格 |数值运算 |数值运算 |... |绝对值
TableForm[data, TableSpacing -> {1, 5},
      |表格形式 |表格空格
      TableHeadings -> {None, {"展开级数n", "在x=1处展开的幂级数的值", "f(1.01)", "Error(%)"}},
      |表格标头 |无
Plot[Evaluate[Table[Normal[Series[f[x], {x, 1, n}]], {n, 5}]],
      |绘图 |计算 |表格 |转换... |级数
      {x, -4, 6}, PlotLegends -> "Expressions"]
      |绘图的图例

```

Out[]:=TableForm=

展开级数n	在x=1处展开的幂级数的值	f(1.01)	Error(%)
1	1.45232	1.4513	0.0699699
2	1.45132	1.4513	0.00133528
3	1.4513	1.4513	0.0000147268
4	1.4513	1.4513	9.99409×10^{-8}
5	1.4513	1.4513	2.24461×10^{-10}
6	1.4513	1.4513	4.10031×10^{-12}
7	1.4513	1.4513	6.11986×10^{-14}
8	1.4513	1.4513	0.
9	1.4513	1.4513	0.

Out[]:=

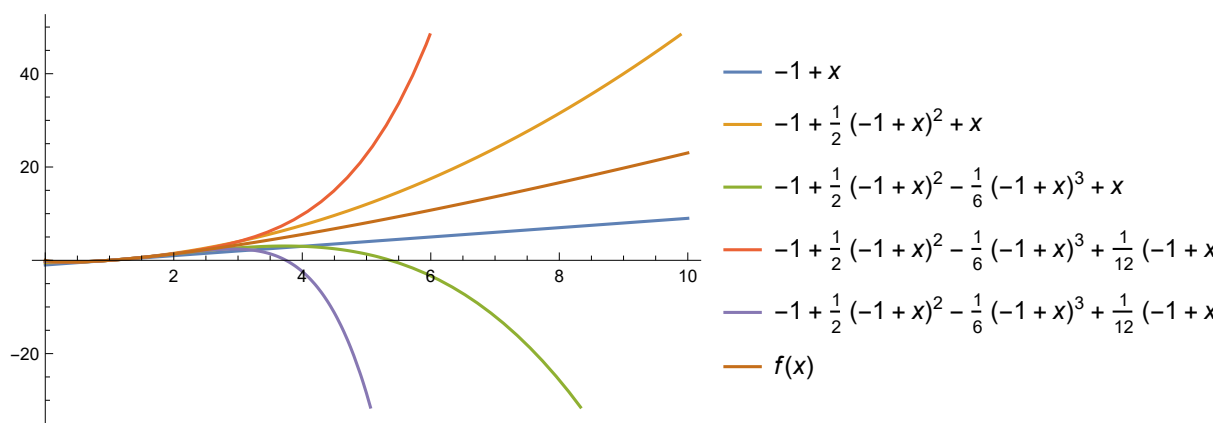


```

In[ ]:= (*3*)
Clear["`*"]
清除
(*在x=1处展开*)
f[x_] := x Log[x]
对数
Plot[{Evaluate[Table[Normal[Series[f[x], {x, 1, n}]], {n, 5}]], f[x]],
绘图 计算 表格 转换... 级数
{x, 0, 10}, PlotLegends -> "Expressions"]
绘图的图例

```

Out[]:=



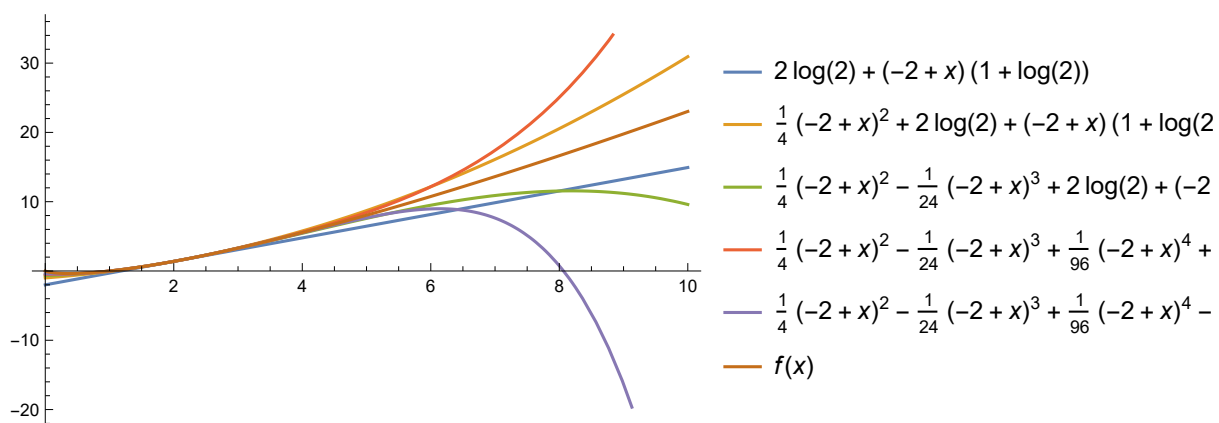
(*在x=2处展开*)

```

In[ ]:= Plot[{Evaluate[Table[Normal[Series[f[x], {x, 2, n}]], {n, 5}]], f[x]],
绘图 计算 表格 转换... 级数
{x, 0, 10}, PlotLegends -> "Expressions"]
绘图的图例

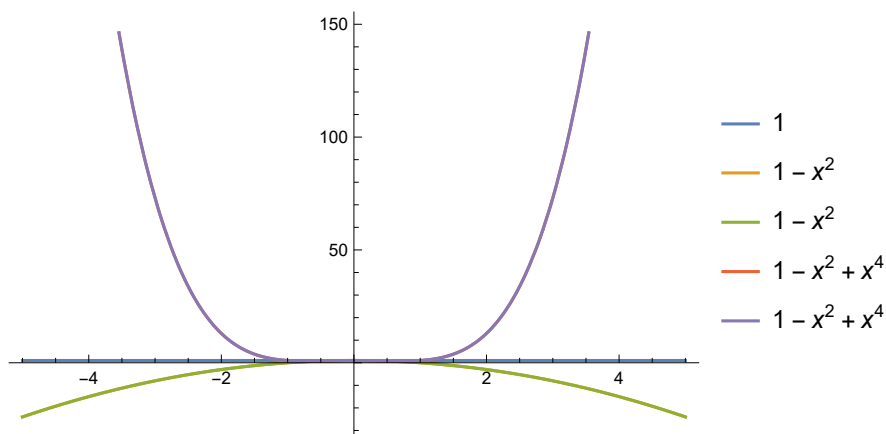
```

Out[]:=



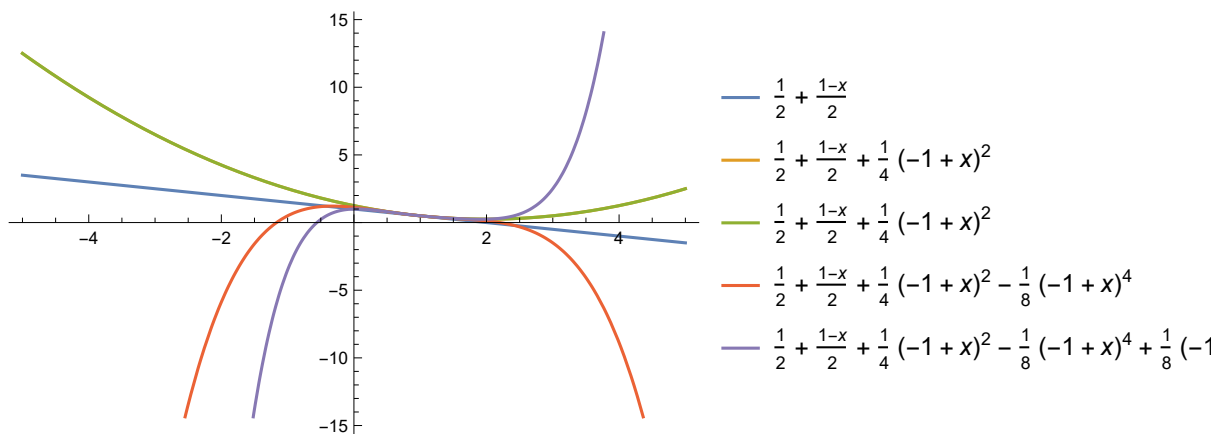

```
In[ ]:= Clear["*"]
清除
(*在x=0处展开*)
f[x_] := 1 / (1 + x^2)
Plot[Evaluate[Table[Normal[Series[f[x], {x, 0, n}]], {n, 5}]],
绘图 计算 表格 转换... 级数
{x, -5, 5}, PlotLegends -> "Expressions"]
绘图的图例
```

Out[]:=



```
In[ ]:= (*在x=1处展开*)
Plot[Evaluate[Table[Normal[Series[f[x], {x, 1, n}]], {n, 5}]],
绘图 计算 表格 转换... 级数
{x, -5, 5}, PlotLegends -> "Expressions"]
绘图的图例
```

Out[]:=

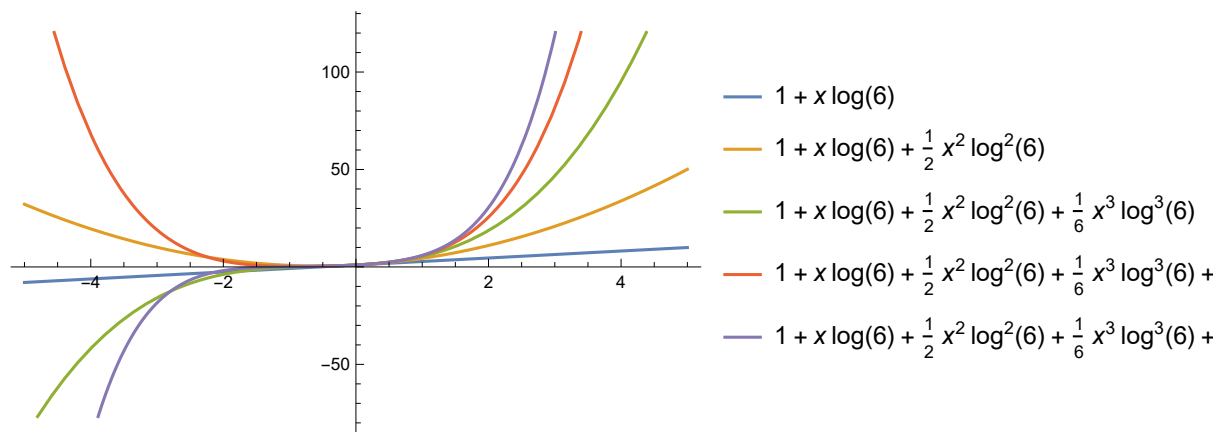


```

In[ ]:= Clear["`*"]
清除
(*在x=0处展开*)
f[x_] := 6^x;
Plot[Evaluate[Table[Normal[Series[f[x], {x, 0, n}]], {n, 5}]],
绘图 计算 表格 转换... 级数
{x, -5, 5}, PlotLegends -> "Expressions"]
绘图的图例

```

Out[]:=

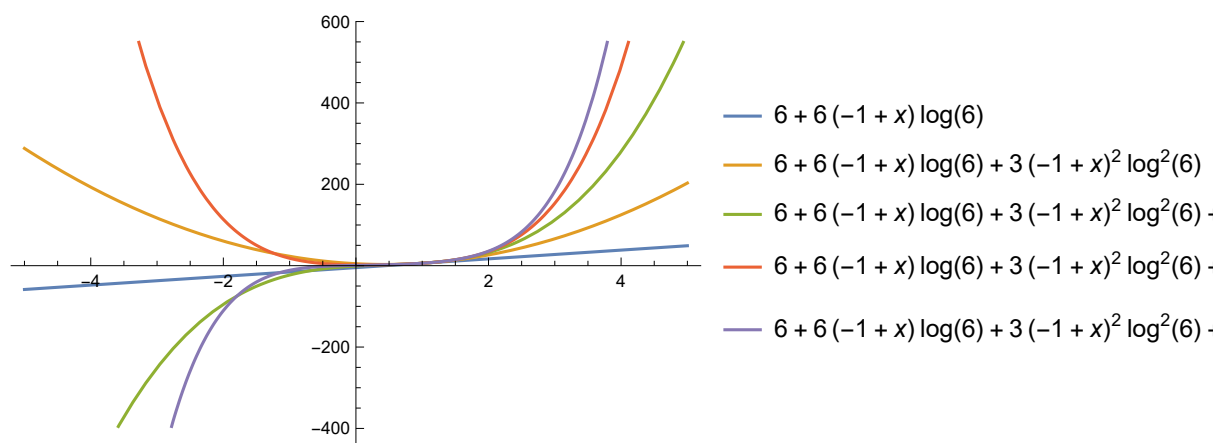


```

In[ ]:= (*在x=1处展开*)
Plot[Evaluate[Table[Normal[Series[f[x], {x, 1, n}]], {n, 5}]],
绘图 计算 表格 转换... 级数
{x, -5, 5}, PlotLegends -> "Expressions"]
绘图的图例

```

Out[]:=

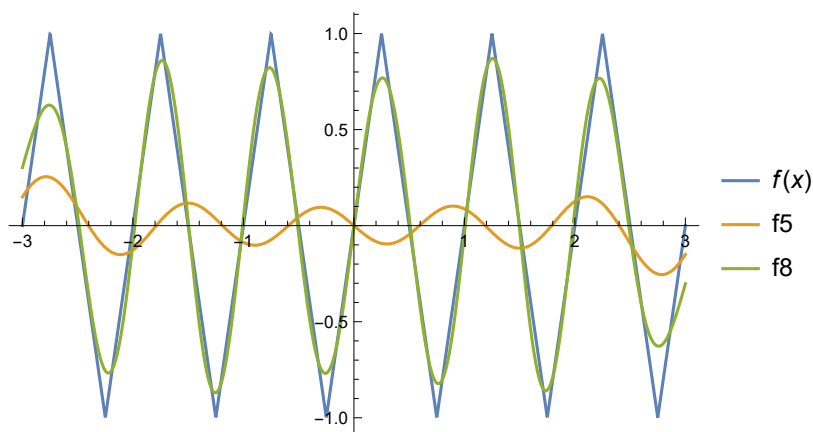


```

In[ ]:= (*4*)
Clear["`*"]
清除
f[x_] := TriangleWave[x];
三角波
f5 = FourierSeries[f[x], x, 5];
傅立叶级数
f8 = FourierSeries[f[x], x, 8];
傅立叶级数
Plot[{f[x], f5, f8}, {x, -3, 3}, PlotLegends -> "Expressions"]
绘图 绘图的图例

```

Out[]:=



```

In[ ]:= F = f[0.2];
value[n_] := FourierSeries[f[x], x, n] /. x -> 0.2
傅立叶级数
data = Table[{n, N[value[n]], N[F], N[Abs[value[n] - F] / F]}, {n, 4, 15, 2}];
表格 数值运算 数... ..绝对值
TableForm[data, TableSpacing -> {1, 5},
表格形式 表格空格
TableHeadings -> {None, {"傅里叶级数", "x=0.2的级数值", "f (0.2) ", "Error"}}]
表格标头 无

```

Out[]:=TableForm=

傅里叶级数	x=0.2的级数值	f (0.2)	Error
4	$-63.1296 + 0. \text{ i}$	0.961538	66.6548
6	$-289.498 + 7.32747 \times 10^{-15} \text{ i}$	0.961538	302.078
8	$-388.847 + 0. \text{ i}$	0.961538	405.401
10	$10618.4 + 2.27374 \times 10^{-13} \text{ i}$	0.961538	11042.2
12	$165869. + 2.34701 \times 10^{-13} \text{ i}$	0.961538	172503.
14	$1.67809 \times 10^6 - 3.6593 \times 10^{-12} \text{ i}$	0.961538	1.74521×10^6

```
In[ ]:= f = FourierTransform[f[x], x, ω]
```

[傅立叶变换](#)

```
DiscretePlot[Abs[f], {ω, 0.01, 10}]
```

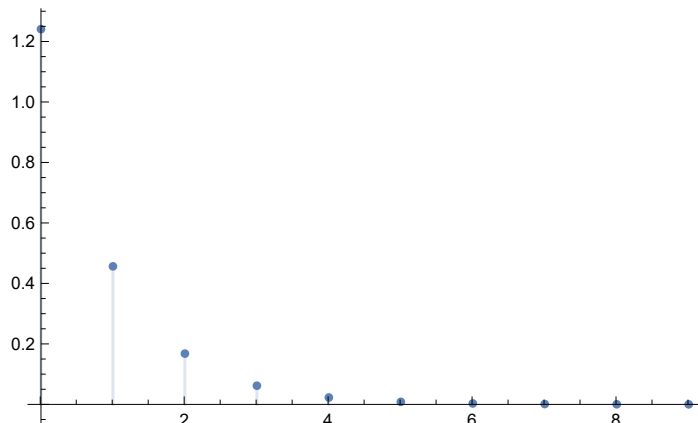
[离散图](#)

[绝对值](#)

```
Out[ ]:=
```

$$e^{-\text{Abs}[\omega]} \sqrt{\frac{\pi}{2}}$$

```
Out[ ]:=
```



```
Clear["'*"]
```

[清除](#)

```
f[x_] := Which[-1/2 ≤ x < 1/2, 1 - x^2]
```

[Which循环](#)

```
f[x_] := f[x - 1] /; x ≥ 1/2
```

```
f[x_] := f[x + 1] /; x ≤ -1/2
```

```
f5 = FourierSeries[f[x], x, 5];
```

[傅立叶级数](#)

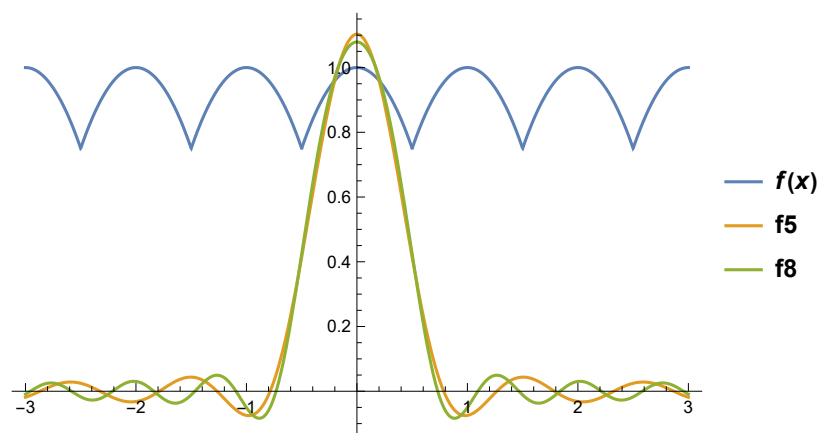
```
f8 = FourierSeries[f[x], x, 8];
```

[傅立叶级数](#)

```
Plot[{f[x], f5, f8}, {x, -3, 3}, PlotLegends → "Expressions"]
```

[绘图](#)

[绘图的图例](#)



```

In[ ]:= F = f[0.2];
value[n_] := FourierSeries[f[x], x, n] /. x -> 0.2
      |傅立叶级数
data = Table[{n, N[value[n]], N[F], N[Abs[value[n] - F] / F]}, {n, 4, 15, 2}];
      |表格 |数值运算 |数... |绝对值
TableForm[data, TableSpacing -> {1, 5},
      |表格形式 |表格空格
      TableHeadings -> {None, {"傅里叶级数", "x=0.2的级数值", "f (0.2)", "Error"}}]
      |表格标头 |无

```

Out[]//TableForm=

傅里叶级数	x=0.2的级数值	f (0.2)	Error
4	$0.915601 + 0. i$	0.96	0.0462489
6	$0.97151 + 0. i$	0.96	0.0119897
8	$0.970474 + 0. i$	0.96	0.0109103
10	$1.00258 + 0. i$	0.96	0.0443552
12	$1.03468 + 0. i$	0.96	0.0777865
14	$1.01186 + 6.93889 \times 10^{-18} i$	0.96	0.0540194

```

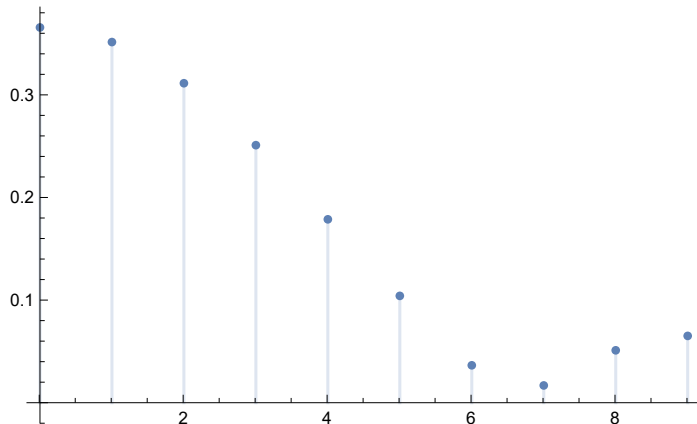
In[ ]:= f = FourierTransform[f[x], x, ω]
      |傅立叶变换
DiscretePlot[Abs[f], {ω, 0.01, 10}]
      |离散图 |绝对值

```

Out[]:=

$$\frac{-4 \omega \cos\left[\frac{\omega}{2}\right] + (8 + 3 \omega^2) \sin\left[\frac{\omega}{2}\right]}{2 \sqrt{2} \pi \omega^3}$$

Out[]:=



```

In[ ]:= Clear["*"]
清除

f[x_] := Piecewise[{{1, 0 ≤ x < 1}, {2 - x, 1 ≤ x ≤ 2}}]
分段函数

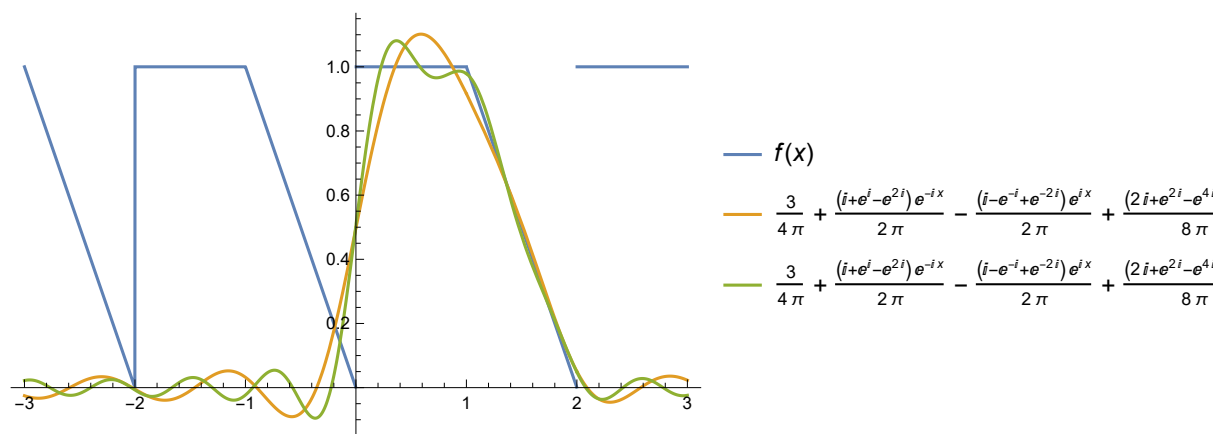
f[x_] := f[x - 2] /; x > 2
f[x_] := f[x + 2] /; x < 0
f5 = FourierSeries[f[x], x, 5];
傅立叶级数

f8 = FourierSeries[f[x], x, 8];
傅立叶级数

Plot[{f[x], Evaluate[f5], Evaluate[f8]}, {x, -3, 3}, PlotLegends → "Expressions"]
绘图 计算 计算 绘图的图例

```

Out[]:=



```

In[ ]:= F = f[0.2];
value[n_] := FourierSeries[f[x], x, n] /. x → 0.2
傅立叶级数

data = Table[{n, N[value[n]], N[F], N[Abs[value[n] - F] / F]}, {n, 4, 15, 2}];
表格 数值运算 数... ..绝对值

TableForm[data, TableSpacing → {1, 5},
表格形式 表格空格

TableHeadings → {None, {"傅里叶级数", "x=0.2的级数值", "f (0.2) ", "Error"}}]
表格标头 无

```

Out[]//TableForm=

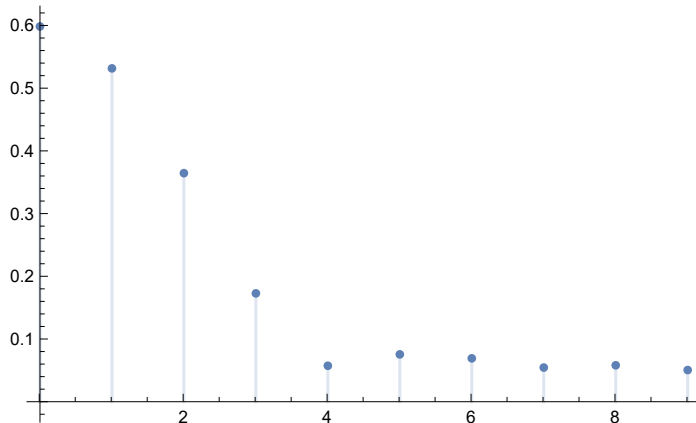
傅里叶级数	x=0.2的级数值	f (0.2)	Error
4	0.763581 + 0. i	1.	0.236419
6	0.872367 + 0. i	1.	0.127633
8	0.961728 + 0. i	1.	0.038272
10	1.0284 - 4.33681 × 10 ⁻¹⁹ i	1.	0.0284039
12	1.06592 + 0. i	1.	0.0659207
14	1.08412 + 0. i	1.	0.0841158

```
In[ ]:= f = FourierTransform[f[x], x, ω]
          傅立叶变换
DiscretePlot[Abs[f], {ω, 0.01, 10}]
          离散图          绝对值
```

Out[]:=

$$\frac{e^{i\omega} - e^{2i\omega} + i\omega}{\sqrt{2\pi}\omega^2}$$

Out[]:=



```
In[ ]:= (*5*)
Clear["`*"]
          清除
f[x_] := Which[-1 ≤ x ≤ 0, -1, 0 ≤ x ≤ 1, 1]
          Which循环
f[x_] := f[x - 2] /; x ≥ 1
f[x_] := f[x + 2] /; x ≤ -1
f8 = FourierSeries[f[x], x, 8]
          傅立叶级数
f11 = FourierSeries[f[x], x, 11]
          傅立叶级数
Plot[{f[x], f8, f11}, {x, -3, 3}, PlotLegends → "Expressions"]
          绘图          绘图的图例
```

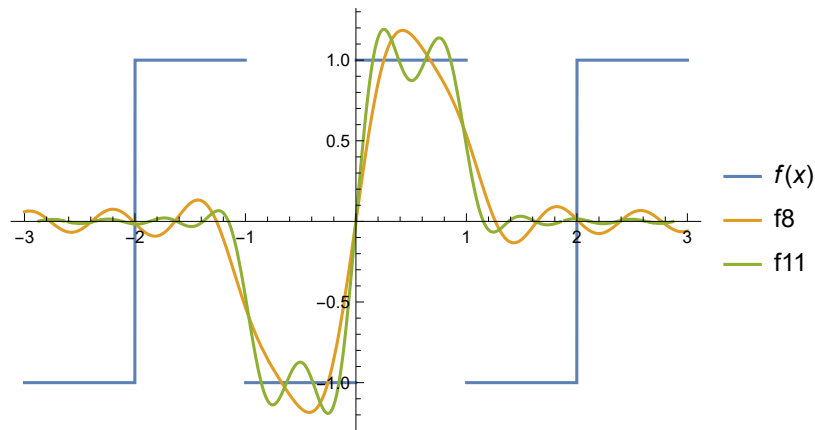
Out[]:=

$$\begin{aligned} & \frac{i(-1+e^i)^2 e^{-i+ix}}{2\pi} + \frac{i(-1+e^{2i})^2 e^{-2i+2ix}}{4\pi} - \frac{i(-1+e^{3i})^2 e^{-3i-3ix}}{6\pi} + \frac{i(-1+e^{3i})^2 e^{-3i+3ix}}{6\pi} - \\ & \frac{i(-1+e^{4i})^2 e^{-4i-4ix}}{8\pi} + \frac{i(-1+e^{4i})^2 e^{-4i+4ix}}{8\pi} - \frac{i(-1+e^{5i})^2 e^{-5i-5ix}}{10\pi} + \\ & \frac{i(-1+e^{5i})^2 e^{-5i+5ix}}{10\pi} - \frac{i(-1+e^{6i})^2 e^{-6i-6ix}}{12\pi} + \frac{i(-1+e^{6i})^2 e^{-6i+6ix}}{12\pi} - \\ & \frac{i(-1+e^{7i})^2 e^{-7i-7ix}}{14\pi} + \frac{i(-1+e^{7i})^2 e^{-7i+7ix}}{14\pi} - \frac{i(-1+e^{8i})^2 e^{-8i-8ix}}{16\pi} + \\ & \frac{i(-1+e^{8i})^2 e^{-8i+8ix}}{16\pi} + \frac{2ie^{-ix} \sin\left[\frac{1}{2}\right]^2}{\pi} + \frac{(-1+e^{2i})e^{-i-2ix} \sin[1]}{2\pi} \end{aligned}$$

Out[*]=

$$\begin{aligned}
& \frac{i \left(-1 + e^i\right)^2 e^{-i + i x}}{2 \pi} + \frac{i \left(-1 + e^{2 i}\right)^2 e^{-2 i + 2 i x}}{4 \pi} - \frac{i \left(-1 + e^{3 i}\right)^2 e^{-3 i + 3 i x}}{6 \pi} + \frac{i \left(-1 + e^{3 i}\right)^2 e^{-3 i + 3 i x}}{6 \pi} - \\
& \frac{i \left(-1 + e^{4 i}\right)^2 e^{-4 i + 4 i x}}{8 \pi} + \frac{i \left(-1 + e^{4 i}\right)^2 e^{-4 i + 4 i x}}{8 \pi} - \frac{i \left(-1 + e^{5 i}\right)^2 e^{-5 i + 5 i x}}{10 \pi} + \\
& \frac{i \left(-1 + e^{5 i}\right)^2 e^{-5 i + 5 i x}}{10 \pi} - \frac{i \left(-1 + e^{6 i}\right)^2 e^{-6 i + 6 i x}}{12 \pi} + \frac{i \left(-1 + e^{6 i}\right)^2 e^{-6 i + 6 i x}}{12 \pi} - \\
& \frac{i \left(-1 + e^{7 i}\right)^2 e^{-7 i + 7 i x}}{14 \pi} + \frac{i \left(-1 + e^{7 i}\right)^2 e^{-7 i + 7 i x}}{14 \pi} - \frac{i \left(-1 + e^{8 i}\right)^2 e^{-8 i + 8 i x}}{16 \pi} + \\
& \frac{i \left(-1 + e^{8 i}\right)^2 e^{-8 i + 8 i x}}{16 \pi} - \frac{i \left(-1 + e^{9 i}\right)^2 e^{-9 i + 9 i x}}{18 \pi} + \frac{i \left(-1 + e^{9 i}\right)^2 e^{-9 i + 9 i x}}{18 \pi} - \\
& \frac{i \left(-1 + e^{10 i}\right)^2 e^{-10 i + 10 i x}}{20 \pi} + \frac{i \left(-1 + e^{10 i}\right)^2 e^{-10 i + 10 i x}}{20 \pi} - \frac{i \left(-1 + e^{11 i}\right)^2 e^{-11 i + 11 i x}}{22 \pi} + \\
& \frac{i \left(-1 + e^{11 i}\right)^2 e^{-11 i + 11 i x}}{22 \pi} + \frac{2 i e^{-i x} \operatorname{Sin}\left[\frac{1}{2}\right]^2}{\pi} + \frac{\left(-1 + e^{2 i}\right) e^{-i - 2 i x} \operatorname{Sin}[1]}{2 \pi}
\end{aligned}$$

Out[*]=

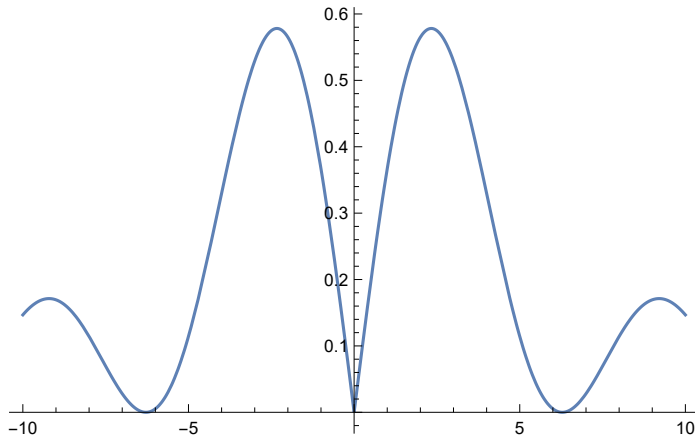



```
In[ ]:= f = FourierTransform[f[x], x, w]
          |傅立叶变换
Plot[Abs[f], {w, -10, 10}, PlotRange -> All]
          |绘图 |绝对值 |绘制范围 |全部
```

Out[]:=

$$\frac{i \sqrt{\frac{2}{\pi}} (-1 + \cos[w])}{w}$$

Out[]:=



```
In[ ]:= Clear["`*"]
          |清除
f[x_] := Which[0 ≤ x ≤ 2, x]
          |Which循环
f[x_] := f[x - 2] /; x ≥ 2
f[x_] := f[x + 2] /; x ≤ 0
f8 = FourierSeries[f[x], x, 8]
      |傅立叶级数
f11 = FourierSeries[f[x], x, 11]
      |傅立叶级数
Plot[{f[x], f8, f11}, {x, -3, 3}, PlotLegends -> "Expressions"]
      |绘图 |绘图的图例
```

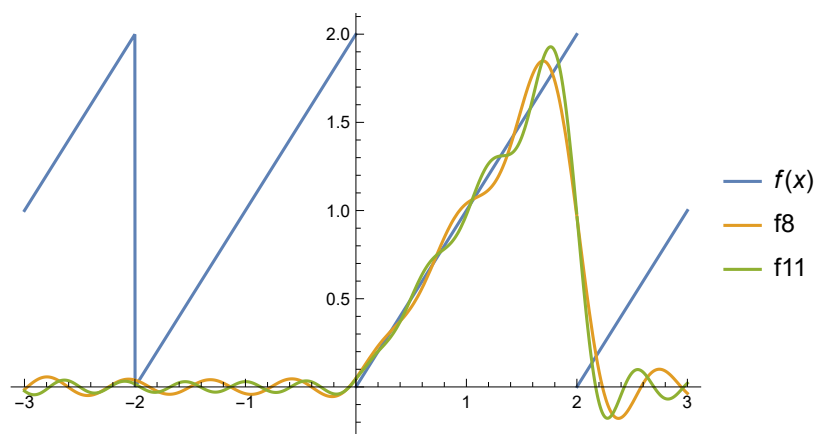
Out[]:=

$$\begin{aligned} & \frac{1}{\pi} - \frac{(1 - (1 - 2i)e^{2i})e^{-ix}}{2\pi} - \frac{(1 - (1 + 2i)e^{-2i})e^{ix}}{2\pi} - \\ & \frac{(1 - (1 - 4i)e^{4i})e^{-2ix}}{8\pi} - \frac{(1 - (1 + 4i)e^{-4i})e^{2ix}}{8\pi} - \frac{(1 - (1 - 6i)e^{6i})e^{-3ix}}{18\pi} - \\ & \frac{(1 - (1 + 6i)e^{-6i})e^{3ix}}{18\pi} - \frac{(1 - (1 - 8i)e^{8i})e^{-4ix}}{32\pi} - \frac{(1 - (1 + 8i)e^{-8i})e^{4ix}}{32\pi} - \\ & \frac{(1 - (1 - 10i)e^{10i})e^{-5ix}}{50\pi} - \frac{(1 - (1 + 10i)e^{-10i})e^{5ix}}{50\pi} - \\ & \frac{(1 - (1 - 12i)e^{12i})e^{-6ix}}{72\pi} - \frac{(1 - (1 + 12i)e^{-12i})e^{6ix}}{72\pi} - \frac{(1 - (1 - 14i)e^{14i})e^{-7ix}}{98\pi} - \\ & \frac{(1 - (1 + 14i)e^{-14i})e^{7ix}}{98\pi} - \frac{(1 - (1 - 16i)e^{16i})e^{-8ix}}{128\pi} - \frac{(1 - (1 + 16i)e^{-16i})e^{8ix}}{128\pi} \end{aligned}$$

Out[*]=

$$\begin{aligned}
& \frac{1}{\pi} - \frac{(1 - (1 - 2i)e^{2i})e^{-ix}}{2\pi} - \frac{(1 - (1 + 2i)e^{-2i})e^{ix}}{2\pi} - \\
& \frac{(1 - (1 - 4i)e^{4i})e^{-2ix}}{8\pi} - \frac{(1 - (1 + 4i)e^{-4i})e^{2ix}}{8\pi} - \frac{(1 - (1 - 6i)e^{6i})e^{-3ix}}{18\pi} - \\
& \frac{(1 - (1 + 6i)e^{-6i})e^{3ix}}{18\pi} - \frac{(1 - (1 - 8i)e^{8i})e^{-4ix}}{32\pi} - \frac{(1 - (1 + 8i)e^{-8i})e^{4ix}}{32\pi} - \\
& \frac{(1 - (1 - 10i)e^{10i})e^{-5ix}}{50\pi} - \frac{(1 - (1 + 10i)e^{-10i})e^{5ix}}{50\pi} - \frac{(1 - (1 - 12i)e^{12i})e^{-6ix}}{72\pi} - \\
& \frac{(1 - (1 + 12i)e^{-12i})e^{6ix}}{72\pi} - \frac{(1 - (1 - 14i)e^{14i})e^{-7ix}}{98\pi} - \frac{(1 - (1 + 14i)e^{-14i})e^{7ix}}{98\pi} - \\
& \frac{(1 - (1 - 16i)e^{16i})e^{-8ix}}{128\pi} - \frac{(1 - (1 + 16i)e^{-16i})e^{8ix}}{128\pi} - \frac{(1 - (1 - 18i)e^{18i})e^{-9ix}}{162\pi} - \\
& \frac{(1 - (1 + 18i)e^{-18i})e^{9ix}}{162\pi} - \frac{(1 - (1 - 20i)e^{20i})e^{-10ix}}{200\pi} - \\
& \frac{(1 - (1 + 20i)e^{-20i})e^{10ix}}{200\pi} - \frac{(1 - (1 - 22i)e^{22i})e^{-11ix}}{242\pi} - \frac{(1 - (1 + 22i)e^{-22i})e^{11ix}}{242\pi}
\end{aligned}$$

Out[*]=



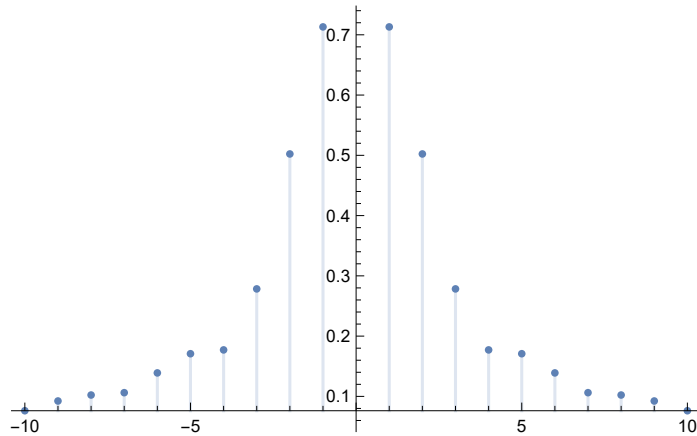
In[*]:= **f = FourierTransform[f[x], x, w]**
 [傅立叶变换]

Out[*]=

$$\text{FourierTransform}\left[\frac{-1 + e^{2iw}(1 - 2iw)}{\sqrt{2\pi}w^2}, x, w\right]$$

In[*]:= **DiscretePlot[Abs[f], {w, -10, 10}, PlotRange -> All]**
 [离散图] [绝对值] [绘制范围] [全部]

Out[*]=



(*6*)

In[*]:= Clear["`*"]

清除

(*三角波*)

f[t_]:= TriangleWave[t];

三角波

a = FourierSeries[f[t], t, 9]

傅立叶级数

Out[*]=

$$\begin{aligned}
& \frac{1}{4\pi} e^{-4i t} \left(-e^{-i} + e^i + e^{-3i} - e^{3i} - e^{-5i} + e^{5i} + e^{-7i} - e^{7i} - e^{-9i} + e^{9i} + e^{-11i} - e^{11i} - 4i(-3 + \pi) \right) + \\
& \frac{1}{4\pi} e^{4i t} \left(e^{-i} - e^i - e^{-3i} + e^{3i} + e^{-5i} - e^{5i} - e^{-7i} + e^{7i} + e^{-9i} - e^{9i} - e^{-11i} + e^{11i} + 4i(-3 + \pi) \right) + \\
& \frac{1}{16\pi} e^{-8i t} \left(-e^{-2i} + e^{2i} + e^{-6i} - e^{6i} - e^{-10i} + e^{10i} + e^{-14i} - e^{14i} - e^{-18i} + e^{18i} + e^{-22i} - e^{22i} - 8i(-3 + \pi) \right) - \\
& \frac{1}{\pi} 4 e^{-\frac{11i}{4} - i t} \left(-1 + e^{\frac{i}{2}} - e^i + e^{\frac{3i}{2}} - e^{2i} + e^{\frac{5i}{2}} - e^{3i} + e^{\frac{7i}{2}} - e^{4i} + e^{\frac{9i}{2}} - e^{5i} + e^{\frac{11i}{2}} - i e^{\frac{11i}{4}}(-3 + \pi) \right) + \\
& \frac{1}{\pi} 4 e^{-\frac{11i}{4} + i t} \left(-1 + e^{\frac{i}{2}} - e^i + e^{\frac{3i}{2}} - e^{2i} + e^{\frac{5i}{2}} - e^{3i} + e^{\frac{7i}{2}} - e^{4i} + e^{\frac{9i}{2}} - e^{5i} + e^{\frac{11i}{2}} - i e^{\frac{11i}{4}}(-3 + \pi) \right) - \\
& \frac{1}{\pi} e^{-\frac{11i}{2} - 2i t} \left(-1 + e^i - e^{2i} + e^{3i} - e^{4i} + e^{5i} - e^{6i} + e^{7i} - e^{8i} + e^{9i} - e^{10i} + e^{11i} + 2i e^{\frac{11i}{2}}(-3 + \pi) \right) + \frac{1}{\pi} \\
& e^{-\frac{11i}{2} + 2i t} \left(-1 + e^i - e^{2i} + e^{3i} - e^{4i} + e^{5i} - e^{6i} + e^{7i} - e^{8i} + e^{9i} - e^{10i} + e^{11i} + 2i e^{\frac{11i}{2}}(-3 + \pi) \right) - \\
& \frac{1}{9\pi} 4 e^{-\frac{33i}{4} - 3i t} \left(-1 + e^{\frac{3i}{2}} - e^{3i} + e^{\frac{9i}{2}} - e^{6i} + e^{\frac{15i}{2}} - e^{9i} + e^{\frac{21i}{2}} - e^{12i} + e^{\frac{27i}{2}} - e^{15i} + e^{\frac{33i}{2}} - 3i e^{\frac{33i}{4}}(-3 + \pi) \right) + \frac{1}{9\pi} 4 e^{-\frac{33i}{4} + 3i t} \\
& \left(-1 + e^{\frac{3i}{2}} - e^{3i} + e^{\frac{9i}{2}} - e^{6i} + e^{\frac{15i}{2}} - e^{9i} + e^{\frac{21i}{2}} - e^{12i} + e^{\frac{27i}{2}} - e^{15i} + e^{\frac{33i}{2}} - 3i e^{\frac{33i}{4}}(-3 + \pi) \right) - \\
& \frac{1}{25\pi} 4 e^{-\frac{55i}{4} - 5i t} \left(-1 + e^{\frac{5i}{2}} - e^{5i} + e^{\frac{15i}{2}} - e^{10i} + e^{\frac{25i}{2}} - e^{15i} + e^{\frac{35i}{2}} - e^{20i} + \right.
\end{aligned}$$

$$\begin{aligned}
& \left(e^{\frac{45i}{2}} - e^{25i} + e^{\frac{55i}{2}} - 5i e^{\frac{55i}{4}} (-3 + \pi) \right) + \frac{1}{25\pi} 4 e^{-\frac{55i}{4} + 5it} \\
& \left(-1 + e^{\frac{5i}{2}} - e^{5i} + e^{\frac{15i}{2}} - e^{10i} + e^{\frac{25i}{2}} - e^{15i} + e^{\frac{35i}{2}} - e^{20i} + e^{\frac{45i}{2}} - e^{25i} + e^{\frac{55i}{2}} - 5i e^{\frac{55i}{4}} (-3 + \pi) \right) - \\
& \frac{1}{9\pi} e^{-\frac{33i}{2} - 6it} \left(-1 + e^{3i} - e^{6i} + e^{9i} - e^{12i} + e^{15i} - e^{18i} + e^{21i} - e^{24i} + \right. \\
& \quad \left. e^{27i} - e^{30i} + e^{33i} + 6i e^{\frac{33i}{2}} (-3 + \pi) \right) + \frac{1}{9\pi} e^{-\frac{33i}{2} + 6it} \\
& \left(-1 + e^{3i} - e^{6i} + e^{9i} - e^{12i} + e^{15i} - e^{18i} + e^{21i} - e^{24i} + e^{27i} - e^{30i} + e^{33i} + 6i e^{\frac{33i}{2}} (-3 + \pi) \right) - \\
& \frac{1}{49\pi} 4 e^{-\frac{77i}{4} - 7it} \left(-1 + e^{\frac{7i}{2}} - e^{7i} + e^{\frac{21i}{2}} - e^{14i} + e^{\frac{35i}{2}} - e^{21i} + e^{\frac{49i}{2}} - e^{28i} + \right. \\
& \quad \left. e^{\frac{63i}{2}} - e^{35i} + e^{\frac{77i}{2}} - 7i e^{\frac{77i}{4}} (-3 + \pi) \right) + \frac{1}{49\pi} 4 e^{-\frac{77i}{4} + 7it} \\
& \left(-1 + e^{\frac{7i}{2}} - e^{7i} + e^{\frac{21i}{2}} - e^{14i} + e^{\frac{35i}{2}} - e^{21i} + e^{\frac{49i}{2}} - e^{28i} + e^{\frac{63i}{2}} - e^{35i} + e^{\frac{77i}{2}} - 7i e^{\frac{77i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{16\pi} e^{-22i + 8it} \left(-1 + e^{4i} - e^{8i} + e^{12i} - e^{16i} + e^{20i} - e^{24i} + e^{28i} - \right. \\
& \quad \left. e^{32i} + e^{36i} - e^{40i} + e^{44i} + 8i e^{22i} (-3 + \pi) \right) - \\
& \frac{1}{81\pi} 4 e^{-\frac{99i}{4} - 9it} \left(-1 + e^{\frac{9i}{2}} - e^{9i} + e^{\frac{27i}{2}} - e^{18i} + e^{\frac{45i}{2}} - e^{27i} + e^{\frac{63i}{2}} - e^{36i} + \right. \\
& \quad \left. e^{\frac{81i}{2}} - e^{45i} + e^{\frac{99i}{2}} - 9i e^{\frac{99i}{4}} (-3 + \pi) \right) + \frac{1}{81\pi} 4 e^{-\frac{99i}{4} + 9it} \\
& \left(-1 + e^{\frac{9i}{2}} - e^{9i} + e^{\frac{27i}{2}} - e^{18i} + e^{\frac{45i}{2}} - e^{27i} + e^{\frac{63i}{2}} - e^{36i} + e^{\frac{81i}{2}} - e^{45i} + e^{\frac{99i}{2}} - 9i e^{\frac{99i}{4}} (-3 + \pi) \right) \\
\\
\ln[*] := & \mathbf{x} = \frac{1}{16\pi} e^{-8it} \left(-e^{-2i} + e^{2i} + e^{-6i} - e^{6i} - \right. \\
& \quad \left. e^{-10i} + e^{10i} + e^{-14i} - e^{14i} - e^{-18i} + e^{18i} + e^{-22i} - e^{22i} - 8i (-3 + \pi) \right) - \\
& \frac{1}{\pi} 4 e^{-\frac{11i}{4} - it} \left(-1 + e^{\frac{i}{2}} - e^i + e^{\frac{3i}{2}} - e^{2i} + e^{\frac{5i}{2}} - e^{3i} + e^{\frac{7i}{2}} - e^{4i} + e^{\frac{9i}{2}} - e^{5i} + e^{\frac{11i}{2}} - i e^{\frac{11i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{\pi} 4 e^{-\frac{11i}{4} + it} \left(-1 + e^{\frac{i}{2}} - e^i + e^{\frac{3i}{2}} - e^{2i} + e^{\frac{5i}{2}} - e^{3i} + e^{\frac{7i}{2}} - e^{4i} + e^{\frac{9i}{2}} - e^{5i} + e^{\frac{11i}{2}} - i e^{\frac{11i}{4}} (-3 + \pi) \right) - \\
& \frac{1}{\pi} e^{-\frac{11i}{2} - 2it} \\
& \left(-1 + e^i - e^{2i} + e^{3i} - e^{4i} + e^{5i} - e^{6i} + e^{7i} - e^{8i} + e^{9i} - e^{10i} + e^{11i} + 2i e^{\frac{11i}{2}} (-3 + \pi) \right) + \\
& \frac{1}{\pi} e^{-\frac{11i}{2} + 2it} \left(-1 + e^i - e^{2i} + e^{3i} - e^{4i} + e^{5i} - e^{6i} + e^{7i} - e^{8i} + \right. \\
& \quad \left. e^{9i} - e^{10i} + e^{11i} + 2i e^{\frac{11i}{2}} (-3 + \pi) \right) - \frac{1}{9\pi} 4 e^{-\frac{33i}{4} - 3it} \\
& \left(-1 + e^{\frac{3i}{2}} - e^{3i} + e^{\frac{9i}{2}} - e^{6i} + e^{\frac{15i}{2}} - e^{9i} + e^{\frac{21i}{2}} - e^{12i} + e^{\frac{27i}{2}} - e^{15i} + e^{\frac{33i}{2}} - 3i e^{\frac{33i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{9\pi} 4 e^{-\frac{33i}{4} + 3it} \left(-1 + e^{\frac{3i}{2}} - e^{3i} + e^{\frac{9i}{2}} - e^{6i} + e^{\frac{15i}{2}} - e^{9i} + e^{\frac{21i}{2}} - e^{12i} + \right. \\
& \quad \left. e^{\frac{27i}{2}} - e^{15i} + e^{\frac{33i}{2}} - 3i e^{\frac{33i}{4}} (-3 + \pi) \right) - \frac{1}{9\pi} e^{-\frac{33i}{2} - 6it} \\
& \left(-1 + e^{3i} - e^{6i} + e^{9i} - e^{12i} + e^{15i} - e^{18i} + e^{21i} - e^{24i} + e^{27i} - e^{30i} + e^{33i} + 6i e^{\frac{33i}{2}} (-3 + \pi) \right) +
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{9\pi} e^{-\frac{33i}{2}+6it} \left(-1 + e^{3i} - e^{6i} + e^{9i} - e^{12i} + e^{15i} - e^{18i} + e^{21i} - e^{24i} + \right. \\
& \quad \left. e^{27i} - e^{30i} + e^{33i} + 6i e^{\frac{33i}{2}} (-3 + \pi) \right) - \frac{1}{49\pi} 4 e^{-\frac{77i}{4}-7it} \\
& \quad \left(-1 + e^{\frac{7i}{2}} - e^{7i} + e^{\frac{21i}{2}} - e^{14i} + e^{\frac{35i}{2}} - e^{21i} + e^{\frac{49i}{2}} - e^{28i} + e^{\frac{63i}{2}} - e^{35i} + e^{\frac{77i}{2}} - 7i e^{\frac{77i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{49\pi} 4 e^{-\frac{77i}{4}+7it} \left(-1 + e^{\frac{7i}{2}} - e^{7i} + e^{\frac{21i}{2}} - e^{14i} + e^{\frac{35i}{2}} - e^{21i} + e^{\frac{49i}{2}} - \right. \\
& \quad \left. e^{28i} + e^{\frac{63i}{2}} - e^{35i} + e^{\frac{77i}{2}} - 7i e^{\frac{77i}{4}} (-3 + \pi) \right) + \frac{1}{16\pi} e^{-22i+8it} \\
& \quad \left(-1 + e^{4i} - e^{8i} + e^{12i} - e^{16i} + e^{20i} - e^{24i} + e^{28i} - e^{32i} + e^{36i} - e^{40i} + e^{44i} + 8i e^{22i} (-3 + \pi) \right); \\
y = & \frac{1}{2+4\pi} e^{-4it} \\
& \quad \left(-e^{-i} + e^i - e^{3i} - e^{3i} - e^{5i} + e^{5i} + e^{7i} - e^{7i} - e^{9i} + e^{9i} + e^{11i} - e^{11i} - 4i (-3 + \pi) \right) + \\
& \frac{1}{2+4\pi} e^{4it} \left(e^{-i} - e^i - e^{3i} + e^{3i} + e^{5i} - e^{5i} - e^{7i} + e^{7i} + e^{9i} - e^{9i} - \right. \\
& \quad \left. e^{11i} + e^{11i} + 4i (-3 + \pi) \right) + \frac{1}{16\pi} e^{-8it} \left(-e^{-2i} + e^{2i} + e^{-6i} - e^{6i} - \right. \\
& \quad \left. e^{-10i} + e^{10i} + e^{-14i} - e^{14i} - e^{-18i} + e^{18i} + e^{-22i} - e^{22i} - 8i (-3 + \pi) \right) - \\
& \frac{1}{\pi} 4 e^{-\frac{11i}{4}-it} \left(-1 + e^{\frac{i}{2}} - e^i + e^{\frac{3i}{2}} - e^{2i} + e^{\frac{5i}{2}} - e^{3i} + e^{\frac{7i}{2}} - e^{4i} + e^{\frac{9i}{2}} - e^{5i} + e^{\frac{11i}{2}} - i e^{\frac{11i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{\pi} 4 e^{-\frac{11i}{4}+it} \left(-1 + e^{\frac{i}{2}} - e^i + e^{\frac{3i}{2}} - e^{2i} + e^{\frac{5i}{2}} - e^{3i} + e^{\frac{7i}{2}} - e^{4i} + e^{\frac{9i}{2}} - e^{5i} + e^{\frac{11i}{2}} - i e^{\frac{11i}{4}} (-3 + \pi) \right) - \\
& \frac{1}{\pi} e^{-\frac{11i}{2}-2it} \\
& \quad \left(-1 + e^i - e^{2i} + e^{3i} - e^{4i} + e^{5i} - e^{6i} + e^{7i} - e^{8i} + e^{9i} - e^{10i} + e^{11i} + 2i e^{\frac{11i}{2}} (-3 + \pi) \right) + \\
& \frac{1}{\pi} e^{-\frac{11i}{2}+2it} \left(-1 + e^i - e^{2i} + e^{3i} - e^{4i} + e^{5i} - e^{6i} + e^{7i} - e^{8i} + \right. \\
& \quad \left. e^{9i} - e^{10i} + e^{11i} + 2i e^{\frac{11i}{2}} (-3 + \pi) \right) - \frac{1}{9\pi} 4 e^{-\frac{33i}{4}-3it} \\
& \quad \left(-1 + e^{\frac{3i}{2}} - e^{3i} + e^{\frac{9i}{2}} - e^{6i} + e^{\frac{15i}{2}} - e^{9i} + e^{\frac{21i}{2}} - e^{12i} + e^{\frac{27i}{2}} - e^{15i} + e^{\frac{33i}{2}} - 3i e^{\frac{33i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{9\pi} 4 e^{-\frac{33i}{4}+3it} \left(-1 + e^{\frac{3i}{2}} - e^{3i} + e^{\frac{9i}{2}} - e^{6i} + e^{\frac{15i}{2}} - e^{9i} + e^{\frac{21i}{2}} - e^{12i} + \right. \\
& \quad \left. e^{\frac{27i}{2}} - e^{15i} + e^{\frac{33i}{2}} - 3i e^{\frac{33i}{4}} (-3 + \pi) \right) - \frac{1}{2+25\pi} 4 e^{-\frac{55i}{4}-5it} \\
& \quad \left(-1 + e^{\frac{5i}{2}} - e^{5i} + e^{\frac{15i}{2}} - e^{10i} + e^{\frac{25i}{2}} - e^{15i} + e^{\frac{35i}{2}} - e^{20i} + e^{\frac{45i}{2}} - e^{25i} + e^{\frac{55i}{2}} - 5i e^{\frac{55i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{2+25\pi} 4 e^{-\frac{55i}{4}+5it} \left(-1 + e^{\frac{5i}{2}} - e^{5i} + e^{\frac{15i}{2}} - e^{10i} + e^{\frac{25i}{2}} - e^{15i} + e^{\frac{35i}{2}} - \right. \\
& \quad \left. e^{20i} + e^{\frac{45i}{2}} - e^{25i} + e^{\frac{55i}{2}} - 5i e^{\frac{55i}{4}} (-3 + \pi) \right) - \frac{1}{9\pi} e^{-\frac{33i}{2}-6it} \\
& \quad \left(-1 + e^{3i} - e^{6i} + e^{9i} - e^{12i} + e^{15i} - e^{18i} + e^{21i} - e^{24i} + e^{27i} - e^{30i} + e^{33i} + 6i e^{\frac{33i}{2}} (-3 + \pi) \right) + \\
& \frac{1}{9\pi} e^{-\frac{33i}{2}+6it} \left(-1 + e^{3i} - e^{6i} + e^{9i} - e^{12i} + e^{15i} - e^{18i} + e^{21i} - e^{24i} + \right.
\end{aligned}$$

$$\begin{aligned}
& \left(e^{27i} - e^{30i} + e^{33i} + 6i e^{\frac{33i}{2}} (-3 + \pi) \right) - \frac{1}{49\pi} 4 e^{-\frac{77i}{4} - 7it} \\
& \left(-1 + e^{\frac{7i}{2}} - e^{7i} + e^{\frac{21i}{2}} - e^{14i} + e^{\frac{35i}{2}} - e^{21i} + e^{\frac{49i}{2}} - e^{28i} + e^{\frac{63i}{2}} - e^{35i} + e^{\frac{77i}{2}} - 7i e^{\frac{77i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{49\pi} 4 e^{-\frac{77i}{4} + 7it} \left(-1 + e^{\frac{7i}{2}} - e^{7i} + e^{\frac{21i}{2}} - e^{14i} + e^{\frac{35i}{2}} - e^{21i} + e^{\frac{49i}{2}} - \right. \\
& \left. e^{28i} + e^{\frac{63i}{2}} - e^{35i} + e^{\frac{77i}{2}} - 7i e^{\frac{77i}{4}} (-3 + \pi) \right) + \frac{1}{16\pi} e^{-22i + 8it} \\
& \left(-1 + e^{4i} - e^{8i} + e^{12i} - e^{16i} + e^{20i} - e^{24i} + e^{28i} - e^{32i} + e^{36i} - e^{40i} + e^{44i} + 8i e^{22i} (-3 + \pi) \right) - \\
& \frac{1}{2 * 81\pi} 4 e^{-\frac{99i}{4} - 9it} \left(-1 + e^{\frac{9i}{2}} - e^{9i} + e^{\frac{27i}{2}} - e^{18i} + e^{\frac{45i}{2}} - e^{27i} + e^{\frac{63i}{2}} - \right. \\
& \left. e^{36i} + e^{\frac{81i}{2}} - e^{45i} + e^{\frac{99i}{2}} - 9i e^{\frac{99i}{4}} (-3 + \pi) \right) + \frac{1}{2 * 81\pi} 4 e^{-\frac{99i}{4} + 9it} \\
& \left(-1 + e^{\frac{9i}{2}} - e^{9i} + e^{\frac{27i}{2}} - e^{18i} + e^{\frac{45i}{2}} - e^{27i} + e^{\frac{63i}{2}} - e^{36i} + e^{\frac{81i}{2}} - e^{45i} + e^{\frac{99i}{2}} - 9i e^{\frac{99i}{4}} (-3 + \pi) \right); \\
z = & \frac{2}{4\pi} e^{-4it} \\
& \left(-e^{-i} + e^i + e^{-3i} - e^{3i} - e^{-5i} + e^{5i} + e^{-7i} - e^{7i} - e^{-9i} + e^{9i} + e^{-11i} - e^{11i} - 4i (-3 + \pi) \right) + \frac{2}{4\pi} \\
& e^{4it} \left(e^{-i} - e^i - e^{-3i} + e^{3i} + e^{-5i} - e^{5i} - e^{-7i} + e^{7i} + e^{-9i} - e^{9i} - e^{-11i} + e^{11i} + 4i (-3 + \pi) \right) + \\
& \frac{1}{16\pi} e^{-8it} \left(-e^{-2i} + e^{2i} + e^{-6i} - e^{6i} - e^{-10i} + e^{10i} + e^{-14i} - \right. \\
& \left. e^{14i} - e^{-18i} + e^{18i} + e^{-22i} - e^{22i} - 8i (-3 + \pi) \right) - \\
& \frac{1}{\pi} 4 e^{-\frac{11i}{4} - it} \left(-1 + e^{\frac{i}{2}} - e^i + e^{\frac{3i}{2}} - e^{2i} + e^{\frac{5i}{2}} - e^{3i} + e^{\frac{7i}{2}} - e^{4i} + e^{\frac{9i}{2}} - e^{5i} + e^{\frac{11i}{2}} - i e^{\frac{11i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{\pi} 4 e^{-\frac{11i}{4} + it} \left(-1 + e^{\frac{i}{2}} - e^i + e^{\frac{3i}{2}} - e^{2i} + e^{\frac{5i}{2}} - e^{3i} + e^{\frac{7i}{2}} - e^{4i} + e^{\frac{9i}{2}} - e^{5i} + e^{\frac{11i}{2}} - i e^{\frac{11i}{4}} (-3 + \pi) \right) - \\
& \frac{1}{\pi} e^{-\frac{11i}{2} - 2it} \\
& \left(-1 + e^i - e^{2i} + e^{3i} - e^{4i} + e^{5i} - e^{6i} + e^{7i} - e^{8i} + e^{9i} - e^{10i} + e^{11i} + 2i e^{\frac{11i}{2}} (-3 + \pi) \right) + \\
& \frac{1}{\pi} e^{-\frac{11i}{2} + 2it} \left(-1 + e^i - e^{2i} + e^{3i} - e^{4i} + e^{5i} - e^{6i} + e^{7i} - e^{8i} + \right. \\
& \left. e^{9i} - e^{10i} + e^{11i} + 2i e^{\frac{11i}{2}} (-3 + \pi) \right) - \frac{1}{9\pi} 4 e^{-\frac{33i}{4} - 3it} \\
& \left(-1 + e^{\frac{3i}{2}} - e^{3i} + e^{\frac{9i}{2}} - e^{6i} + e^{\frac{15i}{2}} - e^{9i} + e^{\frac{21i}{2}} - e^{12i} + e^{\frac{27i}{2}} - e^{15i} + e^{\frac{33i}{2}} - 3i e^{\frac{33i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{9\pi} 4 e^{-\frac{33i}{4} + 3it} \left(-1 + e^{\frac{3i}{2}} - e^{3i} + e^{\frac{9i}{2}} - e^{6i} + e^{\frac{15i}{2}} - e^{9i} + e^{\frac{21i}{2}} - e^{12i} + \right. \\
& \left. e^{\frac{27i}{2}} - e^{15i} + e^{\frac{33i}{2}} - 3i e^{\frac{33i}{4}} (-3 + \pi) \right) - \frac{2}{25\pi} 4 e^{-\frac{55i}{4} - 5it} \\
& \left(-1 + e^{\frac{5i}{2}} - e^{5i} + e^{\frac{15i}{2}} - e^{10i} + e^{\frac{25i}{2}} - e^{15i} + e^{\frac{35i}{2}} - e^{20i} + e^{\frac{45i}{2}} - e^{25i} + e^{\frac{55i}{2}} - 5i e^{\frac{55i}{4}} (-3 + \pi) \right) + \\
& \frac{2}{25\pi} 4 e^{-\frac{55i}{4} + 5it} \left(-1 + e^{\frac{5i}{2}} - e^{5i} + e^{\frac{15i}{2}} - e^{10i} + e^{\frac{25i}{2}} - e^{15i} + e^{\frac{35i}{2}} - \right. \\
& \left. e^{20i} + e^{\frac{45i}{2}} - e^{25i} + e^{\frac{55i}{2}} - 5i e^{\frac{55i}{4}} (-3 + \pi) \right) - \frac{1}{9\pi} e^{-\frac{33i}{2} - 6it}
\end{aligned}$$

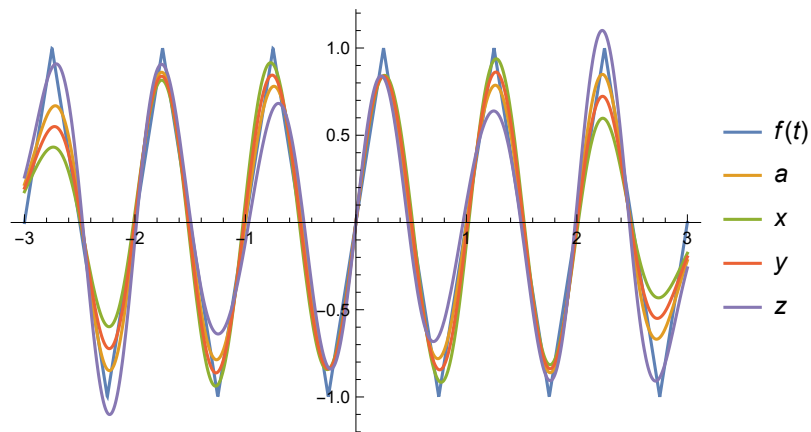
$$\begin{aligned}
& \left(-1 + e^{3i} - e^{6i} + e^{9i} - e^{12i} + e^{15i} - e^{18i} + e^{21i} - e^{24i} + e^{27i} - e^{30i} + e^{33i} + 6i e^{\frac{33i}{2}} (-3 + \pi) \right) + \\
& \frac{1}{9\pi} e^{-\frac{33i}{2} + 6i t} \left(-1 + e^{3i} - e^{6i} + e^{9i} - e^{12i} + e^{15i} - e^{18i} + e^{21i} - e^{24i} + \right. \\
& \quad \left. e^{27i} - e^{30i} + e^{33i} + 6i e^{\frac{33i}{2}} (-3 + \pi) \right) - \frac{1}{49\pi} 4 e^{-\frac{77i}{4} - 7i t} \\
& \left(-1 + e^{\frac{7i}{2}} - e^{7i} + e^{\frac{21i}{2}} - e^{14i} + e^{\frac{35i}{2}} - e^{21i} + e^{\frac{49i}{2}} - e^{28i} + e^{\frac{63i}{2}} - e^{35i} + e^{\frac{77i}{2}} - 7i e^{\frac{77i}{4}} (-3 + \pi) \right) + \\
& \frac{1}{49\pi} 4 e^{-\frac{77i}{4} + 7i t} \left(-1 + e^{\frac{7i}{2}} - e^{7i} + e^{\frac{21i}{2}} - e^{14i} + e^{\frac{35i}{2}} - e^{21i} + e^{\frac{49i}{2}} - \right. \\
& \quad \left. e^{28i} + e^{\frac{63i}{2}} - e^{35i} + e^{\frac{77i}{2}} - 7i e^{\frac{77i}{4}} (-3 + \pi) \right) + \frac{1}{16\pi} e^{-22i + 8i t} \\
& \left(-1 + e^{4i} - e^{8i} + e^{12i} - e^{16i} + e^{20i} - e^{24i} + e^{28i} - e^{32i} + e^{36i} - e^{40i} + e^{44i} + 8i e^{22i} (-3 + \pi) \right) - \\
& \frac{2}{81\pi} 4 e^{-\frac{99i}{4} - 9i t} \left(-1 + e^{\frac{9i}{2}} - e^{9i} + e^{\frac{27i}{2}} - e^{18i} + e^{\frac{45i}{2}} - e^{27i} + e^{\frac{63i}{2}} - \right. \\
& \quad \left. e^{36i} + e^{\frac{81i}{2}} - e^{45i} + e^{\frac{99i}{2}} - 9i e^{\frac{99i}{4}} (-3 + \pi) \right) + \frac{2}{81\pi} 4 e^{-\frac{99i}{4} + 9i t} \\
& \left(-1 + e^{\frac{9i}{2}} - e^{9i} + e^{\frac{27i}{2}} - e^{18i} + e^{\frac{45i}{2}} - e^{27i} + e^{\frac{63i}{2}} - e^{36i} + e^{\frac{81i}{2}} - e^{45i} + e^{\frac{99i}{2}} - 9i e^{\frac{99i}{4}} (-3 + \pi) \right);
\end{aligned}$$

Plot[{f[t], a, x, y, z}, {t, -3, 3}, PlotLegends → "Expressions"]

绘图

绘图的图例

Out[]=



In[]:= **Clear["`*"]**

清除

f[t_] := Which[-1 ≤ t ≤ 0, -1, 0 ≤ t ≤ 1, 1]

Which循环

f[t_] := f[t - 2] /; t ≥ 1

f[t_] := f[t + 2] /; t ≤ -1

w = FourierSeries[f[t], t, 9]

傅立叶级数

Out[]:=

$$\begin{aligned}
 & \frac{i \left(-1 + e^i\right)^2 e^{-i+i t}}{2 \pi} + \frac{i \left(-1 + e^{2 i}\right)^2 e^{-2 i+2 i t}}{4 \pi} - \frac{i \left(-1 + e^{3 i}\right)^2 e^{-3 i-3 i t}}{6 \pi} + \\
 & \frac{i \left(-1 + e^{3 i}\right)^2 e^{-3 i+3 i t}}{6 \pi} - \frac{i \left(-1 + e^{4 i}\right)^2 e^{-4 i-4 i t}}{8 \pi} + \frac{i \left(-1 + e^{4 i}\right)^2 e^{-4 i+4 i t}}{8 \pi} - \\
 & \frac{i \left(-1 + e^{5 i}\right)^2 e^{-5 i-5 i t}}{10 \pi} + \frac{i \left(-1 + e^{5 i}\right)^2 e^{-5 i+5 i t}}{10 \pi} - \frac{i \left(-1 + e^{6 i}\right)^2 e^{-6 i-6 i t}}{12 \pi} + \\
 & \frac{i \left(-1 + e^{6 i}\right)^2 e^{-6 i+6 i t}}{12 \pi} - \frac{i \left(-1 + e^{7 i}\right)^2 e^{-7 i-7 i t}}{14 \pi} + \frac{i \left(-1 + e^{7 i}\right)^2 e^{-7 i+7 i t}}{14 \pi} - \\
 & \frac{i \left(-1 + e^{8 i}\right)^2 e^{-8 i-8 i t}}{16 \pi} + \frac{i \left(-1 + e^{8 i}\right)^2 e^{-8 i+8 i t}}{16 \pi} - \frac{i \left(-1 + e^{9 i}\right)^2 e^{-9 i-9 i t}}{18 \pi} + \\
 & \frac{i \left(-1 + e^{9 i}\right)^2 e^{-9 i+9 i t}}{18 \pi} + \frac{2 i e^{-i t} \operatorname{Sin}\left[\frac{1}{2}\right]^2}{\pi} + \frac{\left(-1 + e^{2 i}\right) e^{-i-2 i t} \operatorname{Sin}[1]}{2 \pi}
 \end{aligned}$$

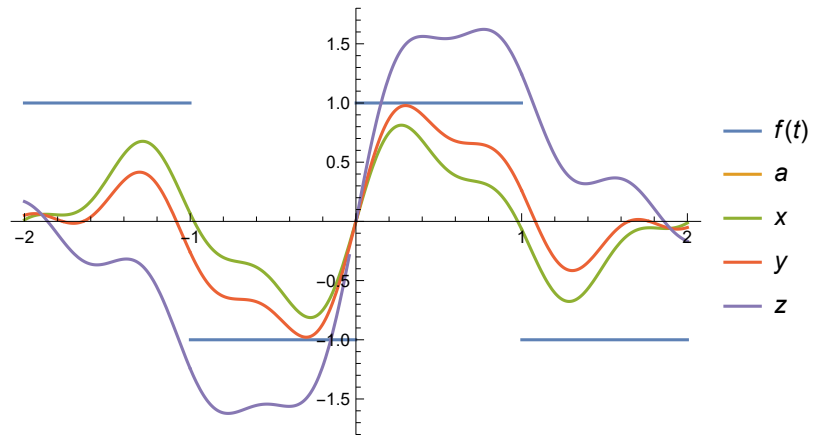
$$\begin{aligned}
\text{In}[*]:= \quad x = & -\frac{i(-1+e^{3i})^2 e^{-3i-3it}}{6\pi} + \frac{i(-1+e^{3i})^2 e^{-3i+3it}}{6\pi} - \frac{i(-1+e^{4i})^2 e^{-4i-4it}}{8\pi} + \\
& \frac{i(-1+e^{4i})^2 e^{-4i+4it}}{8\pi} - \frac{i(-1+e^{5i})^2 e^{-5i-5it}}{10\pi} + \frac{i(-1+e^{5i})^2 e^{-5i+5it}}{10\pi} - \\
& \frac{i(-1+e^{7i})^2 e^{-7i-7it}}{14\pi} + \frac{i(-1+e^{7i})^2 e^{-7i+7it}}{14\pi} - \frac{i(-1+e^{8i})^2 e^{-8i-8it}}{16\pi} + \\
& \frac{i(-1+e^{8i})^2 e^{-8i+8it}}{16\pi} - \frac{i(-1+e^{9i})^2 e^{-9i-9it}}{18\pi} + \frac{i(-1+e^{9i})^2 e^{-9i+9it}}{18\pi}; \\
y = & \frac{i(-1+e^i)^2 e^{-i+it}}{2*2\pi} + \frac{i(-1+e^{2i})^2 e^{-2i+2it}}{2*4\pi} - \frac{i(-1+e^{3i})^2 e^{-3i-3it}}{6\pi} + \\
& \frac{i(-1+e^{3i})^2 e^{-3i+3it}}{6\pi} - \frac{i(-1+e^{4i})^2 e^{-4i-4it}}{8\pi} + \frac{i(-1+e^{4i})^2 e^{-4i+4it}}{8\pi} - \\
& \frac{i(-1+e^{5i})^2 e^{-5i-5it}}{10\pi} + \frac{i(-1+e^{5i})^2 e^{-5i+5it}}{10\pi} - \frac{i(-1+e^{6i})^2 e^{-6i-6it}}{2*12\pi} + \\
& \frac{i(-1+e^{6i})^2 e^{-6i+6it}}{2*12\pi} - \frac{i(-1+e^{7i})^2 e^{-7i-7it}}{14\pi} + \frac{i(-1+e^{7i})^2 e^{-7i+7it}}{14\pi} - \\
& \frac{i(-1+e^{8i})^2 e^{-8i-8it}}{16\pi} + \frac{i(-1+e^{8i})^2 e^{-8i+8it}}{16\pi} - \frac{i(-1+e^{9i})^2 e^{-9i-9it}}{18\pi} + \\
& \frac{i(-1+e^{9i})^2 e^{-9i+9it}}{18\pi} + \frac{2ie^{-it}\text{Sin}\left[\frac{1}{2}\right]^2}{2*\pi} + \frac{(-1+e^{2i})e^{-i-2it}\text{Sin}[1]}{2*2\pi}; \\
z = & \frac{2*i(-1+e^i)^2 e^{-i+it}}{2\pi} + \frac{2*i(-1+e^{2i})^2 e^{-2i+2it}}{4\pi} - \frac{i(-1+e^{3i})^2 e^{-3i-3it}}{6\pi} + \\
& \frac{i(-1+e^{3i})^2 e^{-3i+3it}}{6\pi} - \frac{i(-1+e^{4i})^2 e^{-4i-4it}}{8\pi} + \frac{i(-1+e^{4i})^2 e^{-4i+4it}}{8\pi} - \\
& \frac{i(-1+e^{5i})^2 e^{-5i-5it}}{10\pi} + \frac{i(-1+e^{5i})^2 e^{-5i+5it}}{10\pi} - \frac{2*i(-1+e^{6i})^2 e^{-6i-6it}}{12\pi} + \\
& \frac{2*i(-1+e^{6i})^2 e^{-6i+6it}}{12\pi} - \frac{i(-1+e^{7i})^2 e^{-7i-7it}}{14\pi} + \frac{i(-1+e^{7i})^2 e^{-7i+7it}}{14\pi} - \\
& \frac{i(-1+e^{8i})^2 e^{-8i-8it}}{16\pi} + \frac{i(-1+e^{8i})^2 e^{-8i+8it}}{16\pi} - \frac{i(-1+e^{9i})^2 e^{-9i-9it}}{18\pi} + \\
& \frac{i(-1+e^{9i})^2 e^{-9i+9it}}{18\pi} + \frac{2*2ie^{-it}\text{Sin}\left[\frac{1}{2}\right]^2}{\pi} + \frac{2*(-1+e^{2i})e^{-i-2it}\text{Sin}[1]}{2\pi};
\end{aligned}$$

Plot[{f[t], a, x, y, z}, {t, -2, 2}, PlotLegends → "Expressions"]

绘图

绘图的图例

Out[]=

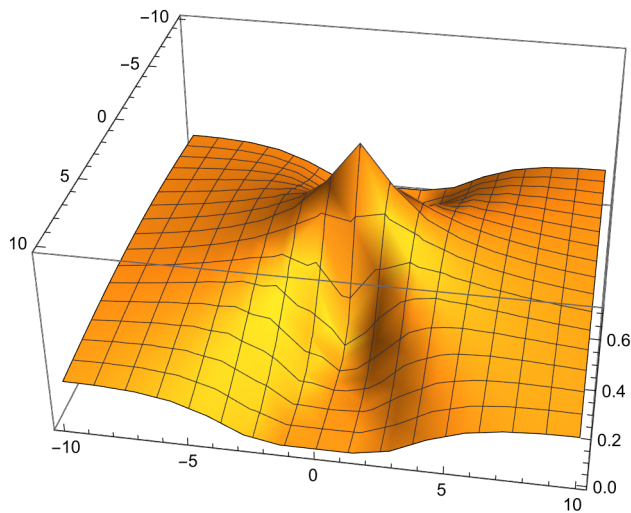


```

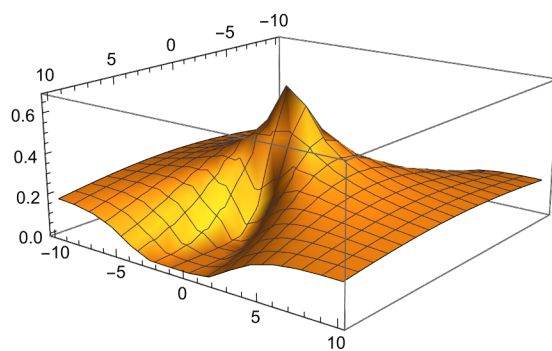
In[ ]:= (*7*)
Clear["`*"]
清除
f1[w_] := Exp[-w^2]
指数形式
f2[w_] := Exp[-w^2 + I w^2]
指数形式 虚数单位
g[w_] := Exp[I 2 w^2 z / 2]
虚数单位
f11[z_, t_] := InverseFourierTransform[f1[w] × g[w], w, t]
傅立叶逆变换
f12[z_, t_] := InverseFourierTransform[f2[w] × g[w], w, t]
傅立叶逆变换
Plot3D[Abs[f11[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
绘制... 绝对值 绘制范围 全部
Plot3D[Abs[f12[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
绘制... 绝对值 绘制范围 全部

```

Out[]:=



Out[]:=

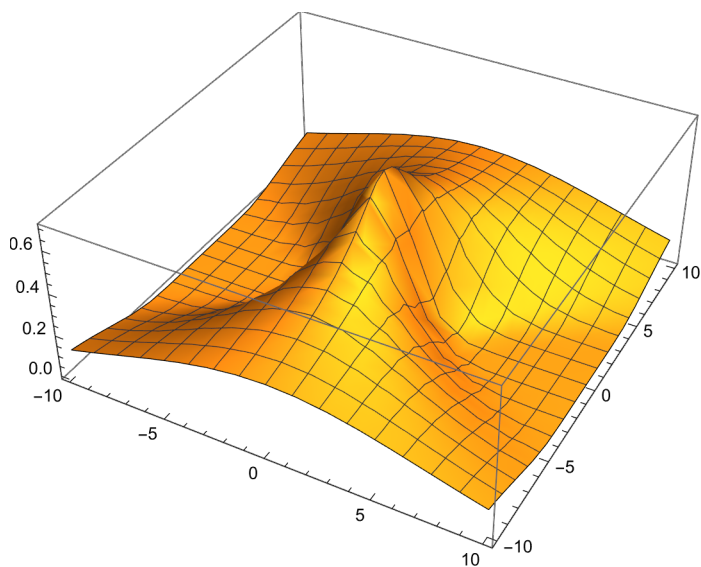


```

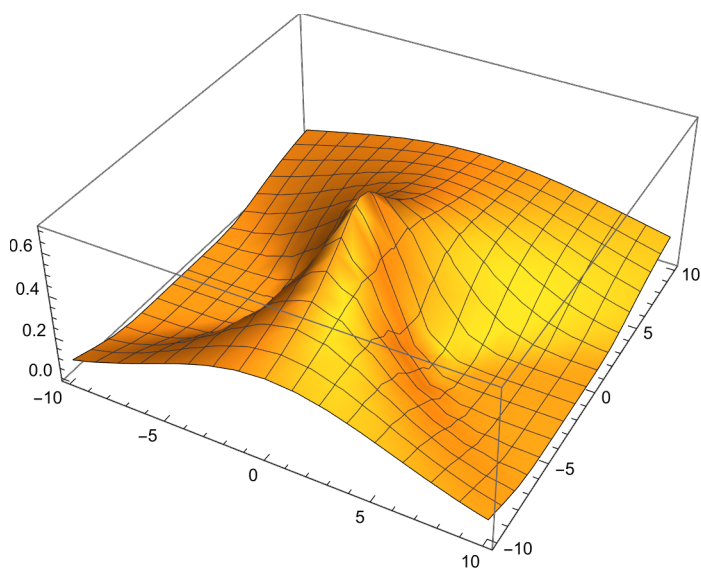
In[ ]:= Clear["*"]
|清除
f1[w_] := Exp[-w^2]
|指数形式
f2[w_] := Exp[-w^2 + I w^2]
|指数形式 |虚数单位
g[w_] := Exp[I 3 w^2 z / Factorial[3]]
|... |虚数单位 |阶乘
f11[z_, t_] := InverseFourierTransform[f1[w] × g[w], w, t]
|傅立叶逆变换
f12[z_, t_] := InverseFourierTransform[f2[w] × g[w], w, t]
|傅立叶逆变换
Plot3D[Abs[f11[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部
Plot3D[Abs[f12[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部

```

Out[]:=



Out[]:=

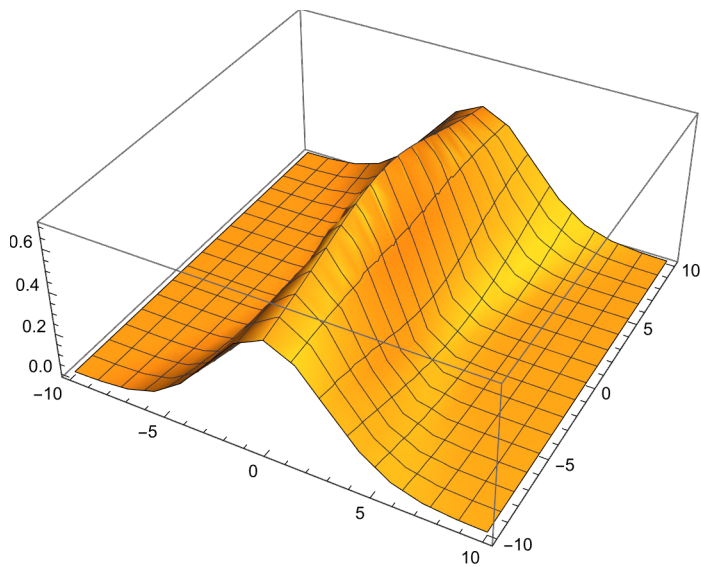


```

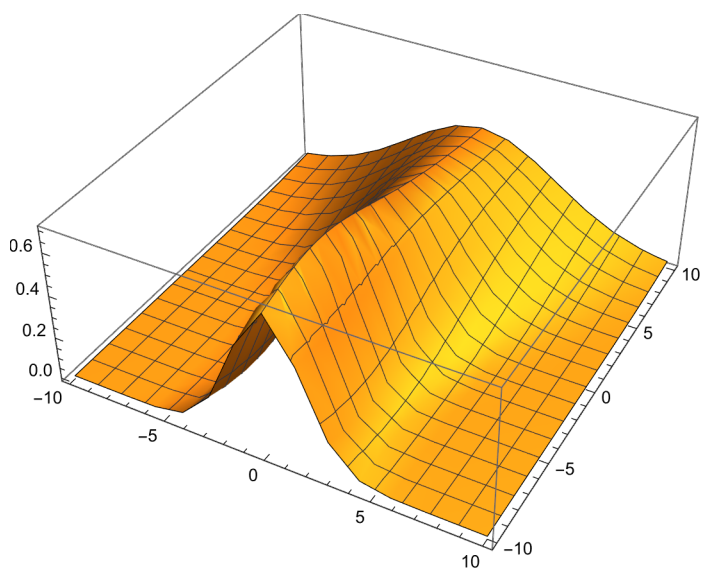
In[ ]:= Clear["`*"]
|清除
f1[w_] := Exp[-w^2]
|指数形式
f2[w_] := Exp[-w^2 + I w^2]
|指数形式 |虚数单位
g[w_] := Exp[I 4 w^2 z / Factorial[4]]
|... |虚数单位 |阶乘
f11[z_, t_] := InverseFourierTransform[f1[w] × g[w], w, t]
|傅立叶逆变换
f12[z_, t_] := InverseFourierTransform[f2[w] × g[w], w, t]
|傅立叶逆变换
Plot3D[Abs[f11[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部
Plot3D[Abs[f12[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部

```

Out[]:=



Out[]:=

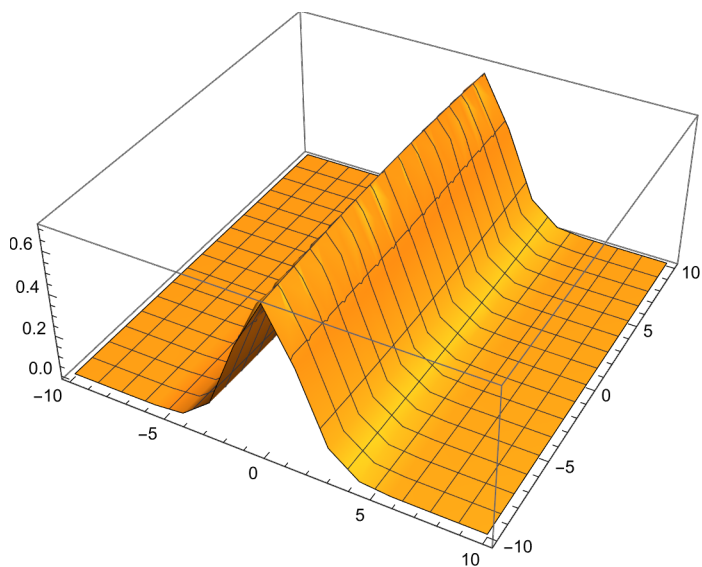


```

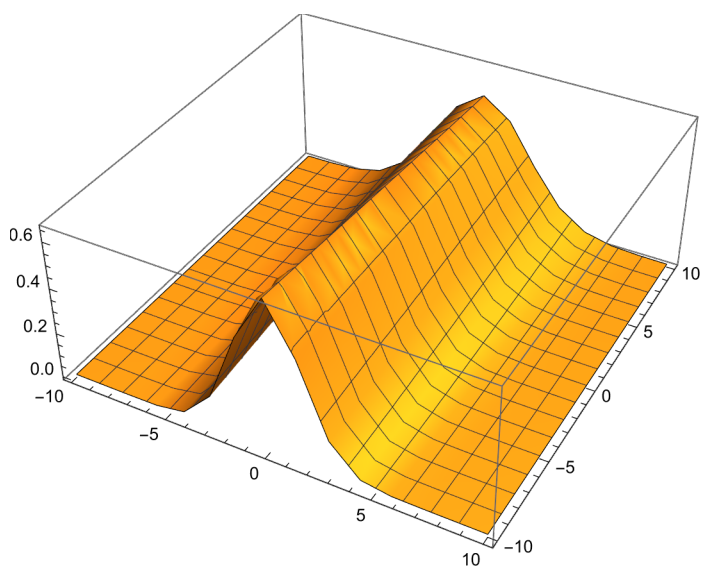
In[ ]:= Clear["`*"]
|清除
f1[w_] := Exp[-w^2]
|指数形式
f2[w_] := Exp[-w^2 + I w^2]
|指数形式 |虚数单位
g[w_] := Exp[I 5 w^2 z / Factorial[5]]
|... |虚数单位 |阶乘
f11[z_, t_] := InverseFourierTransform[f1[w] × g[w], w, t]
|傅立叶逆变换
f12[z_, t_] := InverseFourierTransform[f2[w] × g[w], w, t]
|傅立叶逆变换
Plot3D[Abs[f11[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部
Plot3D[Abs[f12[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部

```

Out[]:=



Out[]:=

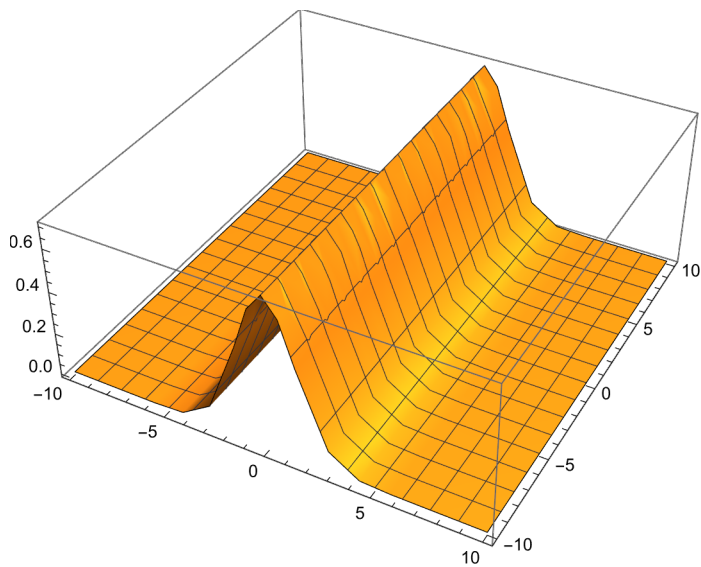


```

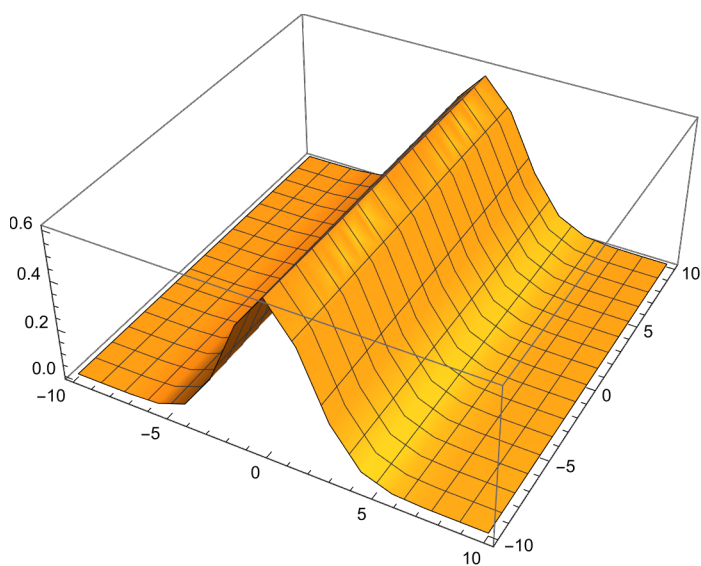
In[ ]:= Clear["`*"]
|清除
f1[w_] := Exp[-w^2]
|指数形式
f2[w_] := Exp[-w^2 + I w^2]
|指数形式 |虚数单位
g[w_] := Exp[I 6 w^2 z / Factorial[6]]
|... |虚数单位 |阶乘
f11[z_, t_] := InverseFourierTransform[f1[w] × g[w], w, t]
|傅立叶逆变换
f12[z_, t_] := InverseFourierTransform[f2[w] × g[w], w, t]
|傅立叶逆变换
Plot3D[Abs[f11[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部
Plot3D[Abs[f12[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部

```

Out[]:=



Out[]:=

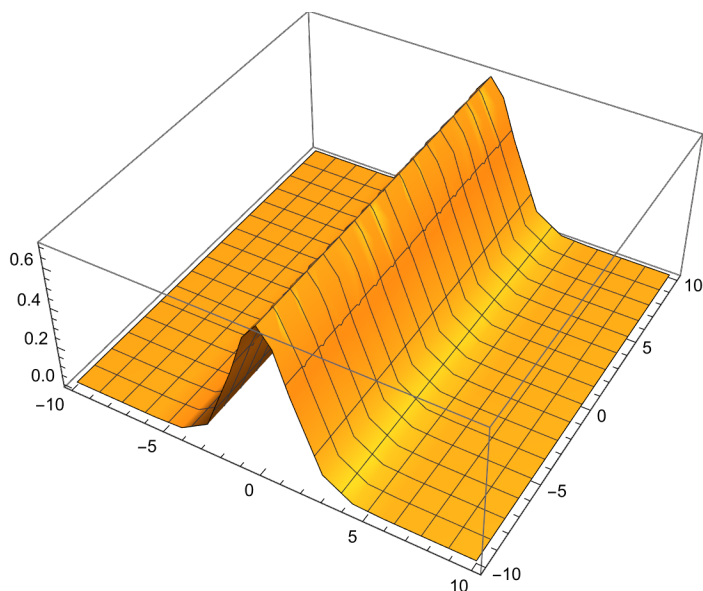


```

In[ ]:= Clear["`*"]
|清除
f1[w_] := Exp[-w^2]
|指数形式
f2[w_] := Exp[-w^2 + I w^2]
|指数形式 |虚数单位
g[w_] := Exp[I 7 w^2 z / Factorial[7]]
|... |虚数单位 |阶乘
f11[z_, t_] := InverseFourierTransform[f1[w] × g[w], w, t]
|傅立叶逆变换
f12[z_, t_] := InverseFourierTransform[f2[w] × g[w], w, t]
|傅立叶逆变换
Plot3D[Abs[f11[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部
Plot3D[Abs[f12[z, t]], {t, -10, 10}, {z, -10, 10}, PlotRange → All]
|绘制... |绝对值 |绘制范围 |全部

```

Out[]:=



Out[]=

